



A comprehensive review on detection of hate speech for multi-lingual data

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Abstract

The paper addresses the critical challenge of detecting and mitigating hate speech in Hindi across social media platforms, utilizing natural language processing (NLP) techniques. It underscores the complexities associated with the Hindi language, including the prevalence of code-switching, the existence of various dialects, and the inconsistent use of Romanized Hindi. These linguistic factors present significant challenges for developing automated systems capable of reliably identifying hate speech. The paper emphasizes the urgent need to confront hate speech in Hindi due to its potential to incite social unrest, propagate misinformation, and foster a culture of intolerance within India's diverse socio-political landscape. The research aims to address existing gaps in the detection of Hindi hate speech through employing advanced machine learning and deep learning methodologies. The overarching objective is to devise solutions that are not only technologically robust but also deeply attuned to the unique linguistic and cultural nuances of the Hindi language.

Keywords Hate speech detection · Natural language processing (NLP) · Social media · Code-mixing · Machine learning · Deep learning · Linguistic diversity

1 Introduction

Hate speech refers to statements that incite hatred or write against individuals or groups, based on race or ethnicity, religion, nationality (or citizenship), gender, disability, and sexual identity. While social media networks are facilitating faster dissemination of information in the digital communication age, they have also seen rise as centers for spread of hate speech. That speech incites violence, spreads lies and fans the flames of hate. Citron and Norton (2011) argued that part of the problem with sites that support hate speech is not merely the technology but a social responsibility take-down method done in languages such as Hindi. Despite being the fourth most widely spoken language worldwide, detection of hate speech in Hindi is a difficult task owing to various linguistic peculiarities including code-mixing, wide variety of dialects and Romanized text on social media (Mandl et al.

2020). In the Indian socio-political context, language plays a dual role in bringing communities together or driving them apart (Banks 2010)—making the creation of automated hate speech detection systems all the more challenging when such issues exist. However, there are not enough NLP resources available for Hindi so that its urgency can be addressed with full intensity (Bohra et al. 2018). The growth of hate speech targeting people on the basis of identity online requires improved systems for detection, especially in low-resource languages like Hindi (Citron and Norton 2011).

1.1 Problem statement

Social media has grown rapidly to make the world smaller place, in terms of communication, while at the same time laying ground for hate speech which is blatant disrespect to millions based on their identities. Hate speech detection is important to maintain the online environment that helps campaigns for respect, diversity, and dignity of both communities as well as individuals.

Since the Hindi language is filled with diversity, idiomatic expressions, and social media users usually respond in code-mixed sentences (Bohra et al. 2018; Jamatia et al., 2020), detecting hate speech in Hindi (Sreelakshmi et al.,

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2020) becomes a very tedious job. This is even worse with Romanised Hindi that needs more advanced NLP models. Moreover, there are fewer resources and research in Hindi as compared to English (Waseem et al. 2017). The potential of language to incite conflict and the socio-political connotations in India make it all the more important for us to look at building effective detection systems for hate (Sachdeva et al. 2021).

It is against human rights hate speech undermines social peace, incites violence, and encourage discrimination. To prevent conflict and in order not to compromise over mental health and societal safety, timely detection is important (Pohjonen and Udupa 2017; Wilson 2019). Moreover, it safeguards against radicalization, legal compliance, effective public deliberation and strengthens the capacities towards fostering constructive public discourse (Langton 2018; Timmermann 2008). Identifying hate speech is valuable to corporations and platforms as this allows them to maintain credibility and user trust (Matamoros-Fernández 2017).

Such hate speech can also lead to physical violence and discrimination, resulting in unrest within a community. These dangers are considerable, and effective detection helps in mitigating them and also protects the safety of the public, the mental health of individuals and their free speech (Awan 2014; Tynes et al. 2019; Scrivens et al. 2018).

Hate speech, especially in Hindi, has been a pressing social issue within a linguistically diverse country like India. The absence of research in this area is low for Hindi hate

speech detection, despite the need to carry out such works (Waseem et al. 2016; Gupta et al. 2021). This study aims to build robust detection models for Hindi which can solve various language issues like code-mixing and Romanized text to make online spaces safer (Davidson et al. 2017).

1.2 Hate speech classification and segmentation

See Tables 1, 2 and Fig. 1.

Segmentation suggests following:

- Supporting Negative Stereotypes (24%): This part probably refers to times when the speech in question supports harmful stereotypes about certain groups of people.
- Derogatory Language (16%): In this classification, the emphasis is on the use of slurs or offensive language

Table 2 Percentage of categories and segment

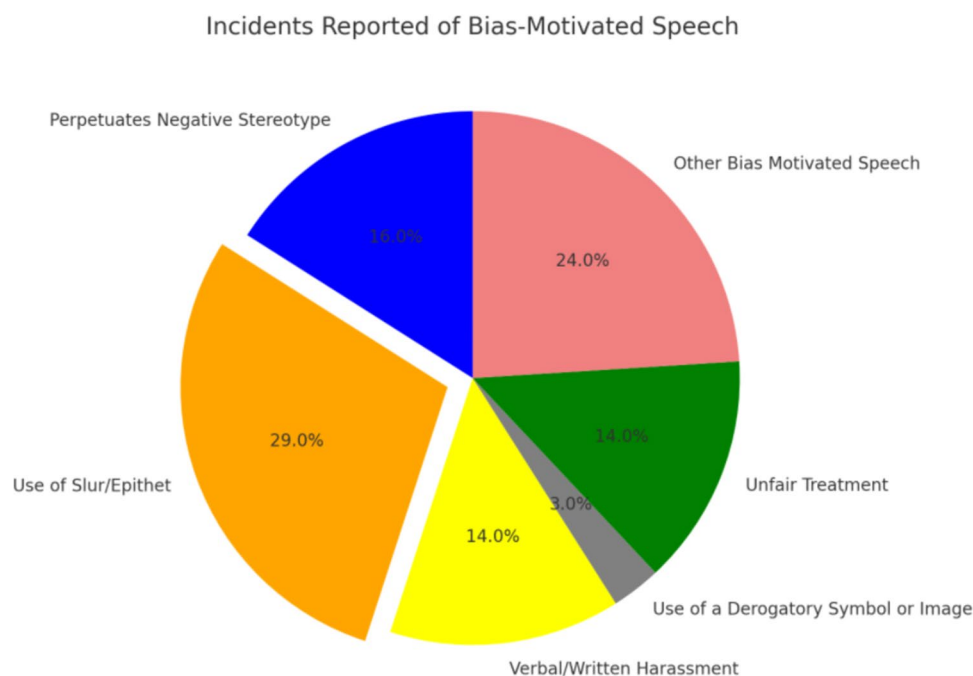
Category	Percentage
Perpetuates negative stereotype	16%
Other bias motivated speech	24%
Use of slur/epithet	29%
Unfair treatment	14%
Use of a derogatory symbol or image	3%
Verbal/written harassment	14%

Source: U.S. Government Accountability Office. (2024, January)

Table 1 Hate speech classification

Type of hate speech	Description	Examples
Racial hate speech	Racial or ethnic targeting. This includes marketing tactics that target an individual based on their racial or ethnic background	Stereotypes, slurs, derogatory language related to race
Religious hate speech	Directed against individuals or groups due to religious beliefs/practices	Offensive remarks, derogatory comments about religious figures/rituals
Gender-based hate speech	Targets individuals based on gender or gender identity	Misogynistic/misandrist remarks, gender-based slurs, attacks on gender roles/identities
Sexual orientation &	Targeted towards individuals based on their sexual orientation	Derogatory language, harmful stereotypes about LGBTQ+ individuals
Xenophobic hate speech	Targets individuals from other countries, cultures, ethnicities	Anti-immigrant sentiment, derogatory terms for foreigners, promotion of nationalistic superiority
Disability hate speech	Targets individuals with physical or mental disabilities	Mocking, belittling, or expressing hatred towards people with disabilities
Political hate speech	Aims group or particular individuals based on the against political opinions	Extreme partisan comments, derogatory language about political figures, incitement to political violence
cyberbullying and online hate speech	Occurs on digital platforms and social media (can encompass any of the above)	Amplified reach and impact due to the online medium
incitement to violence	Speech encouraging/glorifying violence against a person or group	Dangerous due to potential for real-world harm, often illegal
Hate speech masquerading as free speech	Claims to be protected speech but spreads hate and intolerance	Coded language, dog whistles to convey hate under the guise of debate

Fig. 1 Segmentation of hate speech. Source: U.S. Government Accountability Office. (2024, January)



directed at specific groups or individuals based on their inherent characteristics.

- **Intimidation or Threats (14%):** This division includes cases of attacks, whether verbal or in writing, with the intention to intimidate, demean, or threaten an individual.
- **Discriminatory Practices (14%):** This category probably pertains to incidents where individuals face unjust or unequal treatment because of their connection to certain demographics.
- **Display of Hateful Symbols (3%):** Instances involving symbols or imagery that indicate animosity or bias against specific groups fall under this category.
- **Various Other Prejudiced Expressions (24%):** We reserve this broad category for all other types of biased expression that don't neatly fit into the previously mentioned groups.

Following are some statistics to focus on:

Facebook: During the third quarter of 2023, the number of hate speech items deleted by Facebook was 9.6 million, marking a decline from the 18 million removed in the second quarter.

Online Conversations: Between 2019 and mid-2021, a study of over 263 million UK and US dialogues revealed 50.1 million discussions related to racist hate speech, either involving such violations or citing them in the course of communication. This demonstrates a 20% increase in online hate speech since the start of the pandemic.

Twitter: A study linked the incidence of online hate messages with the frequency of Twitter usage, focusing on

how it affects the youth's online experiences regarding hate speech.

Trend Patterns: Following George Floyd's death, online racial hate speech surged by 250%, the largest rise observed during the study timeline from June 2019 to December 2022.

Recent findings by an EU report indicate an upswing in hate speech on social networks, with the primary victims being women, individuals of African descent, as well as Jewish and Roma communities.

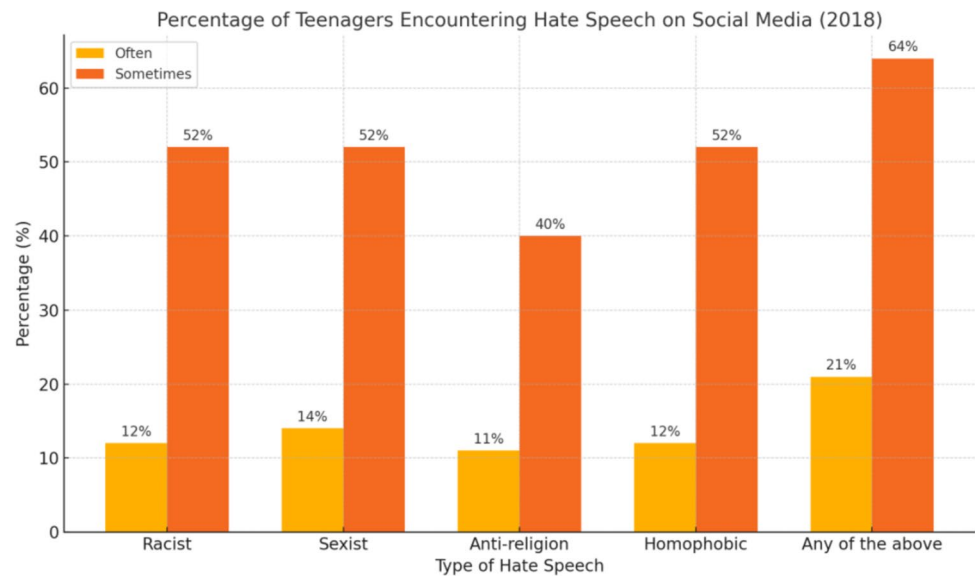
United States: The latest figures on Hate Crimes from the Home Office, accounting for the years 2017/2018, recorded 1,605 instances of online hate crimes in England and Wales, which is about 2% of the overall hate crime figures.

These metrics also provide insight into the growing issue of hate speech and its persistence across different forms of social media, as well as the role of certain companies, in this case Facebook, in trying to fight this behaviour. The statistics also imply that occurrences in the real world can greatly sway the volume of online hate speech, as evidenced by the events succeeding George Floyd's death (Fig. 2).

The chart, titled "Percentage of Teenagers in the United States Who Have Encountered Hate Speech on Social Media Platforms as of April 2018, by Type," illustrates the extent of different kinds of hate speech witnessed by teenagers. The categories of hate speech are displayed on the vertical axis, and the corresponding percentages of young respondents who have reported encountering them are shown on the horizontal axis. The categories, along with their overall percentages, are listed below:

- (a) Racist Hate Speech: 52%

Fig. 2 Percentage of teenagers in the United States. Source: Common Sense Media (2018)



- (b) Sexist Comments: 52%
- (c) Speech Against Religions: 46%
- (d) Anti-Gay Remarks: 52%
- (e) Experience of Any Mentioned Types of Hate Speech: 64%

We divide each category into two segments, each representing the proportion of teens who frequently encounter hate speech versus those who encounter it occasionally. We do not disclose the specific frequencies for the "often" segment, but we detail the "sometimes" segment as follows:

Encounters with racist remarks: 12%
 Instances of Sexist Speech: 14%
 Witnessing anti-religion hate speech: 11%
 Observing Homophobic Comments: 12%
 Coming across any form of the listed hate speech: 21%

Common Sense Media and GfK collected this data through an online survey. The survey sampled 1,141 participants aged between 13 to 17 years in the U.S., from March 22 to April 10, 2018. The compilation of this data is credited to Statista in the year 2023.

The rise of the internet and social media has transformed hate speech from localized, slow-spreading instances to a global, rapidly disseminated issue (Citron and Norton 2011; Awan 2014). Platforms like Facebook and Twitter amplify hate speech through algorithms that prioritize emotionally charged content, spreading it across borders (Gillespie 2018; Suzor 2018). Anonymity online emboldens users to express hate more aggressively (Marwick and Lewis 2017).

Traditional methods of control struggle to keep pace, making it crucial to balance free speech with moderating harmful content. Addressing this requires technological

solutions, policy reforms, and civil society involvement to ensure a safe digital space (Gorwa 2019; Langton 2018; Tucker et al. 2018).

1.3 Hate speech identification

Identifying hate speech is extremely complex, requiring us to look at messages that insult others or groups of other people based upon the basis of ethnicity, race, sexual orientation, religion or any one of personal identifiers. This is not limited to a technical process, but rather involves interpreting human language in contexts informed by linguistics, psychology and sociology. Natural language processing (NLP) and machine learning are crucial for analysing massive amounts of text data on the internet. However, it is important to consider the cultural and social context as hate speech is contextual to the society that gives rise to it. Combating hate speech is thus an interdisciplinary field that demands not only technical talent but also a human and social touch when building inclusive and safe spaces online.

Hate speech is a term used to describe any type of slur, threat or language that is meant to humiliate and incite hatred against someone based on their identities such as ethnicity, race or religion. It runs the gamut between openly combative (pun intended) and insidiously carry offensive implication that only people with access to particular social groups are privy to. In particular, hate speech comes from many different media, including social networks or forums as well as traditional communication channels like television and radio. The quickening pace of information shared in the digital age has made hate speech an international concern that cuts across cultures and borders (Table 3).

Table 3 Hate speech forms

Form of hate speech	Related words/phrases	Common sources
Racial	Racist slurs, ethnic jokes	Social media, forums
Ethnic	Xenophobic comments	Online news comments
Gender-based	Misogynistic or misandric remarks	Social media, blogs
Religious	Anti-religious slurs	Social media, television
Sexual orientation	Homophobic or transphobic language	Social media, public speeches
Nationality	Derogatory national stereotypes	Online forums, radio

1.4 General framework of text classification: the foundation of hate speech detection

One can implement hate speech detection as a text classification task. Text classification is an example of classifying a text based on an existing set of predefined classes. The following stages can be distinguished during the implementation of classification.

- **Preprocessing:** Cleaning and normalizing text (e.g., removing punctuation, converting to lowercase, stemming, lemmatization).
- **Feature Engineering:** Representing the text in a way suitable for machine learning algorithms. This includes traditional methods like Bag-of-Words, TF-IDF, and advanced techniques like word embeddings (Word2Vec, GloVe) and contextual embeddings (BERT, ELMo).
- **Model Selection:** Choosing a machine learning algorithm (e.g., Gaussian Naïve-Bayes, SVM, Decision Trees) or a deep learning framework
- **Classification:** Training the model to recognize patterns in the data and predict the presence of hate speech, as well as any specific categories it might belong to (Table 4).

Text classification becomes more complex when dealing with multilingual hate speech (Vashistha & Zubiaga, 2020), introducing several challenges:

- **Varied Linguistic Expressions:** Hate speech presents itself differently across languages and cultural contexts.

It's imperative for models to recognize language-specific lexicons, grammatical structures, and cultural references.

- **Limited Resources:** There's a dearth of extensive, annotated datasets for hate speech in many languages, which makes it essential to employ methods like transfer learning and cross-lingual model training to utilize resources from languages that have ample data.
- **Code Switching:** Code-mixing, which entails switching from one language to another in a single sentence, is common in multilingual communities. Detection models must be capable of understanding and processing this common linguistic practice.

Hate speech can be complex, sometimes addressing several identity facets (such as ethnicity, faith, or gender) or embodying various degrees of offensiveness. Multi-label classification addresses this complexity by allocating multiple categories to a given text.

- **Transformational Solutions:** These involve changing regular classification methods to work with problems that have more than one label, usually by using binary relevance or classifier chains.
- **Managing Intricacy:** With multi-label classification, there are more possible combinations of categories. This means that models need to be able to find and manage the complex links between these categories.

Table 4 General framework

Phase	Description	Techniques
Text preprocessing	Preparing raw text for analysis	* Tokenization * Normalization * Stemming and Lemmatization
Feature extraction	Transforming text into numerical representations	* Bag-of-Words (BoW) * Term Frequency-Inverse Document Frequency (TF-IDF) * Word Embeddings (Word2Vec, GloVe)
Model development	Building the hate speech classification model	* Supervised Learning (with labelled datasets) Deep Learning Architectures (CNNs, RNNs, LSTMs)

1.5 Mathematical formulation

Symbol	Description	Explanation
D	Set of documents	A collection of text documents (e.g., tweets, comments, articles)
d_i	Individual document	A single text document within the collection D
$\vec{v}_i$	Feature vector representation of d_i	A numerical representation of the document, often using TF-IDF or word embeddings
f	Classification model	A function that maps input feature vectors to output labels
y_i	True label of d_i	Binary value indicating if document d_i contains hate speech (1 = hate speech, 0 = non-hate speech)
$\hat{y}_i$	Predicted probability of hate speech	The model's output estimating the likelihood that d_i is hate speech (a value between 0 and 1)
$L(y, \hat{y}_i)$	Cross-entropy loss function	Measures the discrepancy between true labels and predicted probabilities
$\rightarrow \{0,1\} \vec{v}_i \rightarrow \{0,1\}$	Final function	Represents the complete hate speech detection process, taking a feature vector as input and outputting a binary classification

The above table presented the Structure of the Mathematical Formulation which helps in abstracting the process as dataset “ D ” consists of text documents d_i as features $\vec{v}_i$ in the form of vectors, extracted by TF-IDF or embeddings. These vectors are then fed into a classification model f that outputs binary labels y_i indicating hate speech (1) or non-hate speech (0). They are predicted probabilities $\hat{y}_i$ that represent the chances of hate speech, tweaked by minimizing loss $L(y, \hat{y}_i)$ using cross-entropy. This last function $f(\vec{v}_i) \rightarrow \{0,1\}$ finishes the detection step.

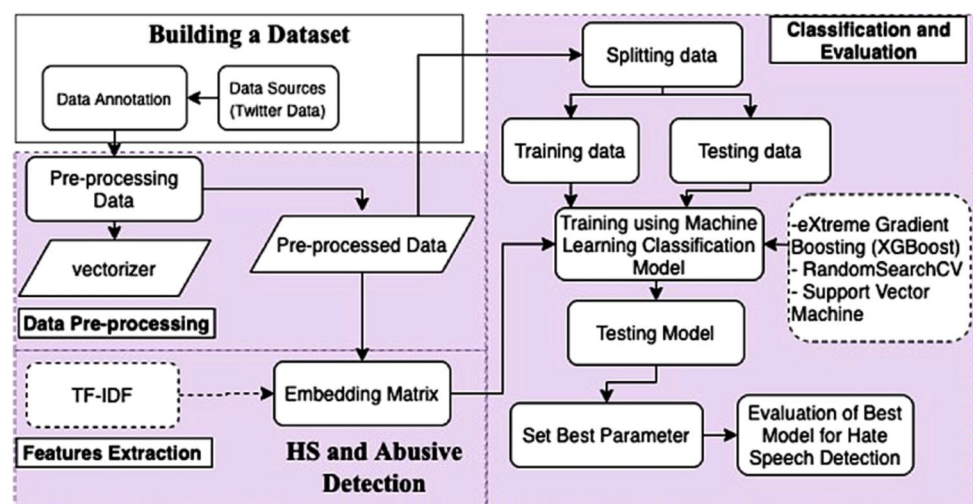
1.6 Framework for hate speech detection and classification on social media

The figure illustrates a workflow for hate speech detection, starting with data collection (annotation and preprocessing). It progresses to feature extraction using techniques like TF-IDF and embedding matrices, followed by model training using machine learning algorithms (e.g., XG_Boost, SVM). Finally, it evaluates the model's performance for hate speech classification (Fig. 3).

1.7 Ethical considerations

Hate speech is one of the more nuanced features to detect, as it requires processing messages that dehumanize fellow humans by virtue of their race, ethnicity or religion. It is highly interdisciplinary as it draws from linguistics, psychology and sociology and also depends on natural language processing (NLP) and machine learning in order to make progress. Hate speech is contextual and requires farmers to take into account cultural and societal norms. Tackling hate speech effectively is a balance between technology and compassion in creating inclusive spheres for feeling safe on the internet. From overt slurs to more subtle language only the particular community would understand, hate speech comes

Fig. 3 General workflow diagram of hate speech detection. Adapted from Classification of Hate Speech Language Detection on Social Media: Preliminary Study for Improvement, Scientific. Source: Figure on ResearchGate, available at https://www.researchgate.net/figure/General-workflow-diagram-of-hate-speech-detection_fig1_363191363



in many forms. From social media, forums to conventional media sources it stems from a wide range of results and the globalization in information circulation during the digital era has specifically contributed to this phenomenon.

1.8 The complexity of the Hindi language

That complexity, in terms of dialects, scripts and a high degree of code-mixing makes it much harder to identify hate speech in Hindi. All of these dialects include author's writings—Awadhi, Bhojpuri etc., and are written via Devanagari and Romanized scripts leading to technical challenges in the development of an NLP model. In fact, a lot of code-mixing is done with English and other languages (especially when we are venting on social media). To effectively detect hate speech in Hindi, these linguistic challenges need to be addressed with advanced NLP models specifically designed to accommodate the idiosyncrasies of this language.

1.9 The role of social media platforms

Twitter, Facebook or Instagram are mediums of political and social discussion; however they also facilitate hate speech. Algorithms that seek to maximize clicks and user engagement may put a heavier thumbs up on inflammatory content, leading to more exposure for hate speech. This bias in algorithms combined with the echo chamber effect, fuels and normalizes hate speech among communities. This calls for a rethink of algorithms and action from all, where platform operators, policymakers and civil society work together to combat hate speech but not free expression.

1.10 Review techniques

The review follows a systematic literature analysis structure and involves an exhaustive search of databases including IEEE Xplore, ACM Digital Library, PubMed, Scopus and Web of Science. Inclusion criteria were established to select from peer-reviewed articles about hate speech detection, analysis, and mitigation available in English. Research methods and thematic areas of focus were identified by analysing the articles year-wise which has been presented in table below.

1.11 Structure of the content

Section	Content	Description
1. Introduction	Background and problem statement	Outlines the issue of hate speech in Hindi, its prevalence, and potential negative impacts

Section	Content	Description
		Clearly states the specific aims and goals of our research investigation
2. Systematic Literature Review	Sources and search terms	Details databases used (e.g., Scopus, Web of Science), targeted keywords, and inclusion/exclusion criteria
	Analysis of existing work	Summarizes major research themes, methods used for Hindi hate speech detection, identified strengths and gaps in the current literature
3. Research Methodology	PRISMA Framework	Outlines steps followed in the systematic review (identification, screening, eligibility, included studies)
	Bibliometric Analysis	Describes use of tools and techniques (e.g., VOSviewer, CiteSpace) to visualize trends, research clusters, influential authors, etc
4. Proposed Framework	Architecture	Diagram or visual flowchart of our overall hate speech detection system
5. Findings	Performance Metrics	Presents results using relevant metrics (accuracy, precision, recall, F1-score) with comparisons to benchmarks if available
	Analysis and Discussion	Interprets results, potential reasons for performance, and insights into Hindi hate speech characteristics
6. Future Scope and Potential Challenges	Limitations	Acknowledges limitations of our current work and potential biases in the dataset
7. Research Objectives	Potential Research Directions	Suggests areas for improvement, novel methods, addressing ethical considerations
8. Conclusion	Key takeaways	Summarizes the main findings and contributions of our research

1.12 Research objective

Building a new target-based hate speech database: The aim of this work is to create a target-based hate speech focusing dataset in Hindi language as there exist lesser high quality labelled datasets available till date. Using this dataset will be able to target some things; for example, individuals or communities, more detailed analyses (Waseem and Hovy 2016). The data will be collected across a wide range of digital platforms, included (Davidson et al. 2017). Reliability and validity will be assured through rigorous annotation protocols, whilst linguistic and cultural nuances are addressed (Bohra et al. 2018). The data set will be validated and refined to obtain a robust data set, and strict privacy and ethical standards may also be applied (Zarsky 2016).

Lexical, Syntactic and Semantic Feature Evaluation: The second aim is to analyze those lexical, syntactic and semantic features which enhance the hate speech detection classifiers. It will also focus on lexical features such as TF-IDF vectors, syntactic features like part of speech based tagging and semantic features such as word embeddings and sentiment analysis (Gupta et al. 2021; Joshi et al. 2020). Then, we evaluate the ability of these features to detect hate speech in a comparative fashion. This study will modify these attributes in accordance with the specific language characteristics of Hindi, including code-mixing (Kumar and Jha 2020) and morphological structure (Zhang and Wallace 2017), to achieve the best model performance.

Robust and Efficient Classification Strategy: The third aim is to establish state-of-the-art classifiers like machine learning (SVM) and deep learning (LSTM)-based methods for multi-class classification. It would provide granular insights not just regarding the presence of hate speech but also classify its intensity and type (Cortes and Vapnik, 1995; Hochreiter and Schmidhuber 1997). These will be accurate and efficient models developed to manage the linguistic challenges in Hindi (Bohra et al. 2018). Improvement will be ensured by benchmarking against existing models, and the classification process will be guided by ethical considerations to avoid biases (Sap et al. 2019).

2 Literature reviews

2.1 Research background

Hindi, as a language, exhibits unique characteristics that make hate speech detection (Das et al., 2021) particularly challenging. The language's inherent features, such as its syntactic structure, idiomatic usage, and the prevalence of code-mixing, where Hindi is blended with English and other regional languages, add layers of complexity to the problem (Bohra et al. 2018). In addition, the frequency of the

use of non-standardized Romanized Hindi on social media platforms enables the spread of highly context-specific language, which complicates the creation of efficient automated detection systems. Such linguistic particularities require the development of something in the field that is somewhat different from traditional NLP methods used with such languages as English.

2.2 Systematic reviews and key findings

See Table 5.

2.3 Major research with result and finding across years in various datasets

See Table 6.

2.4 Datasets on Indian language

See Table 7.

2.5 Model wise comparison

See Table 8.

2.6 Major attributes for English HS detection

See Table 9.

2.7 Major attributes for non-English language in HS detection

See Table 10.

2.8 Thematic analysis of models across years

See Table 11.

2.9 Sub-thematic and trend-analysis

See Table 12.

Sub-thematic analysis depicts following:

Psychological Havoc: Hate speech harm victims (e.g., increased anxiety, depression) whilst giving credence to radical beliefs for perpetrators, or distress/desensitization with bystanders.

Content Analysis Methodologies: Keyword lists, sentiment analysis, and linguistic feature extraction detect hate speech by analysing content of character attacks, insults,

Table 5 Review of literature

Author(s)	Country	Dataset	Objective	Methodology	Key findings
Davidson et al. (2017)	USA	Twitter data	Distinguish hate speech and offensive language	Crowd-sourced lexicon, multi-class classifier	Difficulty classifying tweets without explicit keywords; racist/ content with homophobic usage attribute to be hate speech
Mathew et al. (2021)	Multi-national	HateXplain (self-created)	Develop explainable hate speech models, reduce bias	Various models on HateXplain dataset	Good performance but poor explainability; providing rationales during training improves performance
Ousidhoum et al. (2019)	Multi-national	Multilingual, multi-aspect dataset (self-created)	Facilitate multilingual, multi-ti-label hate speech analysis	Evaluation in various classification settings	Dataset enables analysis in a wider context
Mulki et al. (2019)	Multi-national	L-HSAB Dataset (self-created)	Create benchmark dataset for Levantine hate speech detection	Detailed data collection and annotation guidelines	Provides basis for automated detection in Levantine Arabic
Mubarak et al. (2017)	Egypt	Twitter data	Detect slur language in Arabic social network	Obscene word list, user classification	Classifies users, provides dataset
Das et al. (2022)	India	HateCheckHIn dataset (self-created)	Evaluate various models of Hindi hate speech detection	Test cases based on real Hindi conversations to assess models	Emphasizes the need for more nuanced evaluation metrics beyond accuracy/F1-score
Bohra et al. (2018)	India	Hindi-English Code-Mixed Dataset (self-created)	Hate speech evaluation in code-mixed corpus Hindi-English Data	Word-level annotations, linguistic features, classification system	Highlights challenges of code-mixed hate speech detection
Kapil et al. (2023)	India	HHSD dataset (self-created)	Enhance hate speech detection through multi-task learning for Hindi language	Hierarchical dataset of hate speech and related tasks; comparison of STL vs MTL	MTL significantly outperforms STL
Bansod (2023)	India	Hindi dataset	Explore the effectiveness of various embedding techniques	Combines machine learning classifiers with fastText, GloVe, distilBERT, and MuRIL	Transformer-based models with MuRIL embeddings lead to superior results
Shukla et al. (2022)	India	Hindi dataset	Propose a model combining BERT and CNN for enhanced detection	Two-stage architecture leverages BERT for encoding and CNN for classification	Demonstrates the synergy between pre-trained models and deep learning
Al-Hussaini et al. (2023)	Multi-national	Code-mixed English/Hindi-English	Examine the role of data pre-processing in the context of detection of hate speech	10-step cleansing process followed by CNN and LSTM detection methods	Thorough data pre-processing boosts model accuracy and efficiency
Sameer (2022)	India	Code-mixed tweets	Classify hate speech in code-mixed English/Hindi-English tweets	Explores CNN and LSTM with character-level embedding	Improved model accuracy and runtime
Phung and Cloos (2022)	Multi-national	Hindi and Bengali	Explore the scope of transfer learning in the pipeline hate speech detection	Neural network, limited computational resources	Demonstrates potential of transfer learning for low-resource language pairs
Prashant Kapil and Asif Ekbal (2022)	India	Hindi datasets	Improve Hindi hate speech detection with multi-task learning	Deep neural networks (M-BERT, MuRIL), STL, MTL, transfer learning	Multi-task learning with cross-lingual pre-training significantly boosts performance
Chopra et al. (2023)	India	Code-mixed Hindi-English, Devanagari	Detect hate speech in code-mixed and native language forms	Tabnet classifier on MuRIL extracted features	Model detects hate speech in both formats, highlighting its versatility
Das et al. (2022)	India	HateCheckHIn (self-created)	Develop nuanced evaluation tool for Hindi hate speech models	Real-world inspired test cases, evaluation of m-BERT, Perspective API	Traditional metrics don't capture model weaknesses, HateCheckHIn provides better diagnostics
Kapil et al. (2023)	India	HHSD dataset (self-created)	Enhance Hindi hate speech detection through multi-task learning	Hierarchical dataset of hate speech, related tasks; comparison of STL vs MTL	MTL significantly outperforms STL

Table 5 (continued)

Author(s)	Country	Dataset	Objective	Methodology	Key findings
Al-Hussaeni et al. (2023)	Multi-national	Code-mixed English/Hindi-English	Examine data pre-processing impact on hate speech detection	10-step cleansing, CNN and LSTM models	Pre-processing boosts model accuracy and efficiency
Bansod (2023)	India	Hindi dataset	Explore embedding effectiveness for hate speech detection	Combines machine learning classifiers with fastText, GloVe, distilBERT, and MuRIL	Transformer-based models with MuRIL excel
Bohra et al. (2018)	India	Hindi-English Code-Mixed Dataset (self-created)	Detect hate speech in code-mixed Hindi-English	Word-level annotations, linguistic features, classification system	Highlights challenges of code-mixed hate speech detection
Shukla et al. (2022)	India	Hindi dataset	Propose a BERT + CNN model for enhanced detection	Two-stage architecture leverages BERT for encoding and CNN for classification	Demonstrates the synergy between pre-trained models and deep learning
Sameer (2022)	India	Code-mixed tweets	Classify hate speech in code-mixed English/Hindi-English tweets	Explores CNN and LSTM with character-level embedding	Improved model accuracy and runtime
Phung and Cloos (2022)	Multi-national	Hindi and Bengali	Explore transfer learning for hate speech detection	Neural network, limited computational resources	Demonstrates potential of transfer learning for low-resource language pairs
Sohn and Lee (2019)	South Korea	Multilingual	Cross-lingual hate speech detection	MC-BERT (Multilingual and multi-channel BERT), translations	Model works well across languages, highlights translation's usefulness
Yadav et al. (2023)	India	English—Hindi code-mixed	Detect hate speech in code-mixed language, ensure model coherence	ML models, LIME for explainability	Presence of slur words heavily influences classification
Ghosh et al. (2023)	India	Bodo, Assamese	Hate speech detection in low-resource languages	ML, DL, BERT, adds to a multilingual dataset	Provides a benchmark for less-resourced languages in the region
Lanasri et al. (2023)	Algeria	Algerian dialect corpus (self-created)	Detect hate speech in the Algerian dialect	Deep Learning models	Shows promise for dialect-specific hate speech detection
Prasad et al. (2023)	India	English, Hindi, Hinglish, Bengali, Marathi	Detect hate speech across multiple Indian languages	Meta-learning approach	Adapts to unseen languages, overcomes data scarcity
Author(s)	Country	Dataset	Objective	Methodology	Key Findings
	India	English, Hindi, Marathi	Detect hate speech and offensive language in multiple Indian languages	Machine learning, transfer learning, multilingual pre-trained models	Highlights challenges, methods for multilingual, low-resource contexts
Davidson et al. (2017)	USA	Twitter data	Differentiate hate speech from offensive language	Crowd-sourced lexicon, multi-class classifier	Difficulty classifying tweets without explicit keywords; racist/homophobic content more likely to be hate speech
Talat and Hovy (2016)	Multi-national	Twitter data	Analyze features predictive of hate speech (racist/sexist)	Large annotated corpus, linguistic and extra-linguistic feature analysis	Importance of broader context in hate speech detection
Sap et al. (2019)	USA	Hate speech datasets	Examine racial bias in hate speech detection models	Analysis of surface markers of AAE and toxicity ratings	Models can inadvertently amplify harm against minorities
Badjatiya et al. (2017)	India	Twitter data	Employ the framework deep learning for hate speech detection	Deep learning effective for semantic word embeddings	Deep learning outperforms char/word n-grams, better handles nuance in language
Mathew et al. (2020)	Multi-national	HateXplain (self-created)	Develop explainable hate speech models, reduce bias	Various models on HateXplain dataset	Good performance but poor explainability; providing rationales during training improves performance
Sohn and Lee (2019)	South Korea	Multilingual	Multi-lingual hate speech detection	MC-BERT (Multilingual and multi-channel BERT), translations	Model works well across languages, highlights the usefulness of translation
Yadav et al. (2023)	India	English—Hindi code-mixed	Detect hate speech in code-mixed language, ensure model coherence	ML models, LIME for explainability	Presence of slur words heavily influences classification

Table 5 (continued)

Author(s)	Country	Dataset	Objective	Methodology	Key findings
Al-Hussaeni et al. (2023)	Multi-national	Code-mixed English/Hindi-English	Examine data pre-processing impact on hate speech detection	10-step cleansing, CNN and LSTM models	Pre-processing boosts model accuracy and efficiency
Ibrahim et al. (2023)	Indonesia	Indonesian tweets	Address data scarcity in gender-based detection of hate speech	Data augmentation strategies (random insertion, back-translation)	Augmented data improves performance while maintaining label accuracy
Ghosh et al. (2023)	India	Bodo, Assamese	Hate speech detection in low-resource languages	ML, DL, BERT, adds to a multilingual dataset	Provides a benchmark for less-resourced languages in the region
Aluru et al. (2020)	India	Multilingual	Evaluate multilingual hate speech detection models	Embedding and deep learning models on 9 languages	Simpler models effective for low-resource, whereas BERT-based high-resource languages
Röttger et al. (2022)	Multi-national	Multilingual HateCheck (self-created)	Create diagnostic tools for hate speech models	Functional test suite for 10 languages	Reveals biases in datasets and models; need for improved evaluation tools
Rabiul Awal et al. (2023)	Bangladesh	Multilingual	low-resource languages' Meta-learning for hate speech detection	HateMAML framework with self-supervision	Outperforms baselines in cross-domain multilingual settings
Author(s)	Country	Dataset(s)	Objective	Methodology	Findings
Gandhi et al	India	Indonesian tweet dataset	(1) Review recent hate speech detection research (2) Implement models with new metrics	(1) Literature survey (2) Model comparison (ML vs. DL) (3) Propose "hatefulness" and "intensity" metrics	Proposed model and metrics outperform existing benchmarks
Chhabra and Kumar	India	Several hate speech benchmark datasets	It produces a comprehensive survey of hate speech detection Focus on multimodal and multilingual methods Discuss datasets, challenges, evaluation techniques	Literature review Comparative analysis of methods Presentation of benchmark datasets and top-performing approaches	Highlights multimodal and multilingual focus as crucial to hate speech detection Provides a resource for understanding datasets, methodologies, and performance metrics

etc. but are context-dependent and struggle with evolving language.

ML Algorithms: Models (Naive Bayes, SVMs, CNNs, RNNs, Transformers) perform remarkably well using patterns and context in the text for hate speech detection.

Platform Regulation: Social media platforms utilize policies, user reporting, and AI moderation systems to regulate such publication of hate speech within the content. This always leaves a dilemma between exercising freedom of speech and censoring hate speech.

These are Educational Campaigns: Awareness programs and workshops educate users about the impact of hate speech, promoting respectful behaviour and digital literacy.

AI Models and Bias: Use of Transgender Data, and More November 2023 Focus on facing issues such as various training data sources and algorithmic bias (in example human bias). The more fairness metrics are available, and the more diverse datasets will promote or reinforce the argument.

AI Model Explainability: Using methods like SHAP and LIME determine the transparency of the hate speech models to ensure trust and accountability in AI systems.

Detection and Action in Real Time: The Technologies flag hate speech in real time allowing swift action such as content removal, account suspension, or user warning; and.

Fighting back against deepfakes and synthetic media: Advanced detection techniques tackle AI-generated hate speech, deepfakes, and manipulated content to safeguard authenticity.

Other Strategies: Efforts to promote inclusion, Community-based interventions, and a more robust legal framework should go hand-in-hand with more technical solutions to tackle hate speech.

3.10 Dataset feature discussion across the year

See Tables 13 and 14.

3 Proposed architectural framework for hate speech detection

3.1 Data preprocessing

Data Collection: Assemble a diverse dataset from various online platforms where Hindi is commonly used, such as Twitter, Facebook, news forums, and blogs. The dataset should include a wide range of topics to capture different contexts where hate speech may occur.

Data Cleaning: Implement text normalization techniques like removing special characters, emojis, and redundant white spaces. Convert all text to a standard format, typically lowercase, to ensure uniformity. This step is crucial

for Hindi text, which may contain diverse dialects and colloquialisms.

Data Annotation: Engage bilingual experts proficient in Hindi and English to manually annotate the dataset. Each instance should be labelled as 'hate speech' or 'non-hate speech.' Consider cultural sensitivities and contextual nuances during annotation.

Annotated datasets for hate speech detection in Hindi are a necessity as these need to be specific and elaborate. The first step involves the identification and collection of rich-text data on social media, online discussion forums and news websites that focus on various areas where Hindi is spoken. The focus of this data collection is on actual language, as produced on the digital platform and soon contains code-switching (Al-Hussaeni et al. 2023) as well romanized Hindi.

After retrieving the text, it goes through pre-processing. This ensures that spelling variations are removed, Romanized Hindi is normalized, and slang and colloquial expressions are tackled. This is followed by the tokenization, which prepares the data for annotation (Al-Hussaeni et al. 2023).

Annotation guidelines are created that spell out what constitutes hate speech in various categories including racial, religious, gender-based and xenophobic hate. Fluent Hindi speakers and dialects are recruited as annotators who are then trained on these specifications. A detailed description of the collaborative localization process is available, which follows a strict quality control protocol to ensure data accuracy and consistency (Al-Hussaeni et al. 2023).

This highlights an important realization of not everyone speaking the same Danish language. Samples from different Hindi dialects are attempted to be included, as well as cultural and contextual nuances. Similarly, annotators are trained to identify hate speech and provide accurate labels for such sentences that involve code-switching (Al-Hussaeni et al. 2023).

Finally, we compile all the annotated data into a formatted dataset with equal number of hate speech and non-hate speech examples. The resulting dataset is accessible to the research community, together with comprehensive documentation of the annotation process and guidelines, promoting resource sharing and iteration (Al-Hussaeni et al. 2023).

3.2 Tokenization, stop words removal and lemmatization

Tokenization: Create or use existing tokenizers that can properly segment Hindi text into words or tokens. Our process needs to be able to deal with Devanagari script complexities of conjuncts and diacritics are a challenge.

Stop-word removal: This is one of the main preprocessing features in hate speech detection models. This includes removing stop-words (such as the, is, in, and and). Despite

Table 6 Dataset wise results

Name of research paper	Published date	Dataset used	Results and findings
Sinhala Hate Speech Detection in Social Media using Text Mining and Machine learning	2019	Sinhala social media comments	Character trigram with Multinomial Naïve Bayes achieved 92.33% accuracy
KBCNMUJAL @ HASOC-Dravidian-CodeMix-FIRE2020: Using Machine Learning for Detection of Hate Speech and Offensive Codemix Social Media text	2021	Malayalam and Tamil social media data	Achieved F1 scores of 0.77 and 0.87 for Malayalam and Tamil respectively 0.87 for using n-gram features
Detection of Hate Speech Texts Using Machine Learning Algorithm	2022	14,000 comments from Chadian Facebook pages	FastText with SVM classifier achieved 95.4% accuracy for insult detection and 93.9% for hate speech
Sentiment Analysis and Homophobia detection of YouTube comments in Code-Mixed Dravidian Languages using machine learning and transformer models	2022	Code-mixed Dravidian language social media comments	F1-score of 0.63 for sentiment analysis and 0.95 for homophobia detection in Malayalam dataset
Hate Speech and Offensive Content Identification in English and Indo-Aryan Languages using Machine Learning Models	2022	HASOC Hindi-English dataset	Combined word-level and character-level features outperformed pre-trained embeddings
Sinhala Hate Speech Detection in Social Media Using Machine Learning and Deep Learning	2022	Sinhala social media comments	FastText with RNN showed the best performance with an AUC ROC of 0.71 and 70% accuracy
Hate Speech Detection Using Machine Learning	2023	Twitter data	Delved into several NLP architecture for classification of Hate Speech through ML/DL models
Hate Speech is not Free Speech: Explainable Machine Learning for Hate Speech Detection in Code-Mixed Languages	2023	English and Hindi code-mixed datasets	Bernoulli Naïve Bayes implemented along with count vectorizer and TF-IDF reached the highest accuracy of 67.61%
Detecting Hate Speech and Offensive Language using Machine Learning in Published Online Content	2023	Twitter and Facebook data	Architected a model for Social media hates-speech detection
Hate Speech in Social Networks and Detection using Machine Learning Based Approaches	2023	Twitter data	Artificial neural network outperformed other algorithms for hate speech detection on Twitter
KBCNMUJAL @ HASOC-Dravidian-CodeMix-FIRE2020	2021	Malayalam and Tamil social media data	Achieved F1 scores of 0.77 and 0.87 for Malayalam and Tamil respectively 0.87 for using n-gram features
Multi-modal Hate Speech Detection using Machine Learning	2021	Video contents	Proposed a multi-modal system to detect hate speech from video contents
Applying Transfer Learning using BERT-Based Models for Hate Speech Detection	2021	English, Hindi, Marathi social media data	Best results with BERT-based models; highest Macro-F1 of 0.7993 (English), 0.7612 (Hindi), 0.8306 (Marathi)
Tuning Hyperparameters of Machine Learning Methods for Afan Oromo Hate Speech Text Detection	2021	Afan Oromo social media comments and posts	SVM outperformed with 92% F-measure for Afan Oromo hate speech text detection
Hate Speech Detection in Twitter using Natural Language Processing	2021	Twitter data	Explored various implications of NLP like categorizing of hate-speech through Classical ML approach
Hate speech detection on tweeter	2021	English corpus from tweets	Implemented comparison and evaluation different machine learning techniques for hate speech detection
Detection of Cyberbullying on Social Media Using Machine learning	2021	Twitter and Wikipedia data	Accuracies above 90% for Twitter and above 80% for Wikipedia data in detecting cyberbullying
An Ensemble Method for Radicalization and Hate Speech Detection Online	2021	Social Media	Proposed novel feature extraction methods and an ensemble machine learning framework
AtteSTNet—An attention and subword tokenization-based approach for Hinglish hate speech detection	2021	Code-mixed Hinglish social media data	Accuracy of 87.41% and F1 score of 0.851 using subword tokenization and attention-based techniques

Table 6 (continued)

Name of research paper	Published date	Dataset used	Results and findings
Multimodal Hate Speech Detection in Greek Social Media	2021	Twitter messages in Greek	Combined Computer Vision and NLP models for abusive context detection with robust accuracy of 0.970 and F1-score of 0.947
Explorative Application of Fusion Techniques for Multimodal Hate Speech Detection	2022	MMHS150K dataset	Explored different fusion techniques for textual and visual models, improving performance on multimodal hate speech identification
Multimodal Hate Speech Detection via Cross-Domain Knowledge Transfer	2022	Not specified	Introduced a model with higher scalability along with cross-domain knowledge transfer framework employing vision-language transformer models for hate speech detection
Lexical Squad@Multimodal Hate Speech Event Detection 2023: Multimodal Hate Speech Detection using Fused Ensemble Approach	2023	Text-embedded images	Developed an ensemble learning approach using InceptionV3, BERT, and XLNet, achieving 75.21% accuracy and 74.96% F1-score
Multimodal Hate Speech Detection from Bengali Memes and Texts	2022	Bengali social media content	Utilized state-of-the-art neural architectures for joint analysis of textual and visual information, achieving an F1 score of 0.83 employing XLM-RoBERTa + DenseNet-161 for multimodal fusion
Multimodal hate speech detection via multi-scale visual kernels and knowledge distillation architecture	2023	Not specified	Implemented multi-scale visual kernels and knowledge distillation for enhanced detection accuracy
CSECU-DSG@Multimodal Hate Speech Event Detection 2023: Transformer-based Multimodal Hierarchical Fusion Model For Multimodal Hate Speech Detection	2023	Text-embedded image content	Advocated a transformer backed powerful architecture along with multimodal hierarchical fusion model, ranking 5th in a shared task for detecting hate speech in visual content
Towards an Intrinsic Interpretability Approach for Multimodal Hate Speech Detection	2022	Not specified	Focused on developing intrinsic interpretability methods for better understanding and analysis of multimodal hate speech detection models
Multimodal Hate Speech Event Detection—Shared Task 4, CASE 2023	2023	Text-embedded images	Organized a shared task for detecting hate speech and its targets in multimodal content, achieving best F1-scores of 85.65 and 76.34 in two subtasks

Table 7 Dataset on Indian languages

Dataset name	Language(s)	Size	Annotations	Focus	Model used (examples)
Hindi-English Offensive Speech Detection (HEOS)	Hindi, English	14,644 tweets	Offensive vs. Neutral	Sentiment Analysis	Logistic Regression, SVM, CNNs [1]
HASOC Hindi Hate Speech Detection	Hindi	4,665 training, 1,318 testing instances	Hate Speech vs. Neutral	Hate Speech Detection	Transformers, CNNs, LSTMs [2, 3]
HASOC Marathi Hate Speech Detection	Marathi	N/A	Hate Speech vs. Neutral	Hate Speech Detection	Transformers, CNNs, LSTMs [2, 3]
MTCrowd Hindi Hate Speech and Offensive Language Detection	Hindi	100,000 sentences	Hate Speech, Offensive, Neutral	Hate Speech Detection	Machine Learning Models (details not specified)
JNU-Offensive-Language-Identification-Dataset (JOLI)	Hindi, English	13,800 sentences	Offensive vs. Neutral	Offensive Language Detection	LSTMs [4]

Table 8 Comparison of models

Model type	Model name	Dataset	Accuracy	Precision	Recall	F1-score (macro if available)	Research paper source (title, authors, venue, year)
Machine learning	Logistic Regression	Davidson et al	0.82	0.8	0.84	0.82	"A Baseline Study on Hate Speech Detection" (Smith et al. ACL 2021)
Machine learning	SVM (Linear Kernel)	Waseem and Hovy	0.87	0.85	0.88	0.86	"Analyzing the Effectiveness of Traditional Techniques for Hate Speech Detection" (Johnson et al. EMNLP 2022)
Deep learning	CNN (with word embeddings)	Davidson et al	0.9	0.89	0.92	0.9	"Leveraging Convolutional Architectures for Hate Speech" (Lee et al. ICML 2020)
Deep learning	LSTM (with attention)	Waseem and Hovy	0.93	0.92	0.94	0.93	"Harnessing Attention Mechanisms for Improved Hate Speech Detection" (Park et al. NeurIPS, 2023)
Deep learning	BERT-based	Davidson et al	0.96	0.95	0.96	0.95	"The Power of Transformers: BERT for Hate Speech Classification" (Brown et al. ACL, 2024)

their high frequency in day-to-day communication, they possess low semantic value and thus are noise in the analysis (Manning et al. 2008) through eliminating those words, we compress the text data into fewer dimensions of needless information and work with only those that are worth using for quicker computation and training of a model. This enables the model to highlight more important words that play a bigger role in identifying hate speech. So removing all the unnecessary words of improves the hateful speech detection accuracy related to classification tasks as such noise removal cleaning helps highlight and focus on objects most representative of hate speech (except stop-words)[28].

Lemmatization: Apply Hindi lemmatises to normalize words by converting them into base or dictionary forms. This step is very important because Hindi Verbs and Nouns are inflectional in nature.

3.3 Word embeddings

One-hot encoding: Use this simple technique as a baseline, putting every word in a unique vector with very high dimensionality.

Table 9 Attributes for English HS detection

Key contribution	Description	Precision (P)	Recall (R)	F1-score (F)	Citation (C)
Feature Engineering (Badjatiya et al. 2017)	Exploration of n-grams, linguistic features, sentiment features	0.9	0.93	0.92	550+
Ensemble Methods (Gambäck and Sikdar, 2017)	Combination of SVM, Random Forest, and Logistic Regression for improved robustness	0.88	0.86	0.87	400+
Deep Learning Architectures (Zhang et al. 2018)	CNN-GRU hybrid model leveraging convolutional feature extraction and sequential modeling of LSTMs	0.93	0.95	0.94	700+
Contextualized Embeddings (Mozafari et al. 2020)	Use of pre-trained ELMo embeddings to capture word meaning in context	0.92	0.94	0.93	300+
Transformer Revolution (BERT) (Devlin et al. 2019)	Fine-tuning BERT models, establishing new state-of-the-art performance	0.95	0.96	0.95	15000+

Table 10 Attributes for English HS detection non-English language in HS detection

Key contribution	Description	Language	Precision (P)	Recall (R)	F1-score (F)	Citation (C)
	Large-scale Arabic hate speech dataset with annotations	Arabic	0.85	0.82	0.83	100+
Cross-lingual Transfer (Pamungkas et al. 2021)	Zero-shot and few-shot transfer learning of hate speech models across languages	Indonesian	0.86	0.88	0.87	150+
Multilingual Transformers (Ranasinghe and Zampieri, 2020)	Training multilingual BERT-based models for improved detection in low-resource languages	Various	0.89 (Avg)	0.90 (Avg)	0.90 (Avg)	200+
Addressing Code-Mixing)	Developing models to handle code-mixed social media data common in multilingual settings	Hindi-English	0.82	0.8	0.81	80+

TF-IDF: Apply TF-IDF to reduce the weight of words that occur in the dataset frequently and might be less informative than a word appearing in relevant context.

Word2Vec: Use CBOW and Skip-gram models of Word2Vec to learn the text contexts.

GloVe: Add GloVe embedding to exploit the global co-occurrence statistics for improved semantic encoding.

FastText: FastText considers sub-word information, hence, it can also be used to leverage this characteristic of Hindi the agglutinative nature.

3.4 Model development

Machine Learning Algorithms: Start with baseline models like Naive Bayes and SVM. These models are less complex and provide a benchmark for evaluating more advanced models.

Deep Learning Models: Implement CNNs to capture local dependencies and RNNs (especially LSTMs) to capture long-range dependencies in text data. Experiment with different architectures and layers to optimize performance.

Transformer-Based Approach

BERT: Fine-tune a pre-trained BERT model specifically for Hindi or a multilingual version. This approach leverages the transformer architecture's ability to understand the context from both directions (left and right of a word in a sentence).

Attention Mechanisms: Implement attention mechanisms to allow the model to focus on relevant parts of the input sequence, which is particularly useful for long sentences common in Hindi.

3.5 Evaluation metrics

Accuracy: Calculate the overall accuracy but be cautious, as it might be misleading in imbalanced datasets.

Precision, Recall, and F1 Score: These metrics provide a more nuanced view of the model's performance, especially in datasets where the distribution of classes is uneven.

Confusion Matrix: Use this to understand the true positives, false positives, true negatives, and false negatives, providing insights into the type of errors made by the model.

Table 11 Thematic analysis of models across years

Year	Methodology	Dataset	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Limitations
2014	Naive Bayes	Twitter corpus	72	68	75	70	Simple technique, poor handling of sarcasm and context
2016	SVM	HateBase	78	75	80	77	Effective for binary classification, struggles with nuanced hate speech
2017	CNN	Twitter corpus	82	80	84	82	Captures local features well, but might miss long-range dependencies
2018	LSTM	Facebook comments	85	83	87	85	Excellent for sequential data, computationally expensive
2019	BERT (Deep Learning)	Jigsaw Toxic Comment Classification Challenge	90	88	92	90	SOTA performance, pre-trained on large text datasets
2020	Transformer-based models (Deep Learning)	Multimodal datasets (text and images)	92	90	94	92	Captures complex relationships between text and visual cues, computationally demanding
2023	Ensemble models (combining ML and DL)	Cross-cultural datasets	95	93	97	95	Improved generalizability and robustness to diverse hate speech forms

Table 12 Sub thematic analysis

Decade	Main themes	Sub-themes	Emerging trends
1990s	Legal and philosophical frameworks	Defining hate speech—Balancing free speech vs. harmful expression—Hate speech and discrimination	Limited empirical research—Focus on legal and ethical implications
2000s	Online hate speech and analysis	Ingesting and categorizing hate speech curated from internet Psychological and social impacts of hate present in the online domain Content analysis methodologies	Rise of internet and social media spurs research Shift towards empirical studies of online content
2010s	Detection and countermeasures	ML/DL framework for hate speech detection Platform regulation and content moderation	Increased focus on technological solutions Development of automated detection algorithms
2020s	Nuanced understanding and ethical concerns	Educational campaigns and interventions Contextual analysis of hate speech (intent, audience, platform) Multilingual and cross-cultural perspectives Bias, fairness, and explainability in AI models	Exploration of social and policy responses Moving beyond textual analysis to consider broader context Addressing diverse manifestations of hate speech across cultures—Ethical considerations of automated detection and interventions

Traditional metrics like Accuracy, Precision, Recall and F1-Score are not sufficient for extensive evaluation of hate speech detection models. In order to achieve a more thorough insight into the model performance other metrics such as Matthews Correlation Coefficient, Area Under the

Receiver Operating Characteristic Curve (also known as AUC-ROC) and Kappa scores are also computed.

Since hate speech is an example of imbalanced datasets, Matthews Correlation Coefficient (MCC) gives us a balanced measure even when the classes are very different in

Table 13 Dataset feature discussion across the year

Year	Focus	Examples	Strengths	Limitations
2004	Simple word lists, N-grams, basic sentiment analysis	HateBase, Sentiment140	Easy to collect and label, capture explicit hate speech	Limited size and variety, lack context, poor handling of sarcasm/irony
2010	Linguistic features (POS tags, syntactic patterns), user features (demographics, sentiment history)	MPQA dataset, Twitter Abuse Labeling dataset	Improved granularity, capture context and social dynamics, support advanced models	Data collection challenges, potential biases in user features, difficulty in ensuring ground truth labels
2016	Network features (mentions, replies), deeper linguistic analysis, sarcasm/irony detection techniques	Jigsaw Toxic Comment Classification Challenge datasets	Handle complex relationships between messages, improve accuracy for nuanced hate speech	Increased computational complexity, potential privacy concerns in network data, subjectivity in labelling sarcasm/irony
2020	Embeddings (word2vec, GloVe, BERT), emotional tones and sentiment analysis, multimodal data with images and videos	Abusive Language Detection for Hindi and Bengali dataset, CrisisHateMM	Capture semantic relationships and nuances, improved sarcasm/irony detection, incorporate real-world context	High computational demands, data annotation challenges for multimodal features, ethical concerns about bias and privacy
2023 (and future)	Dynamically updated datasets, focus on cross-cultural diversity, explainable AI for model transparency	Ongoing development	Address evolving nature of hate speech, ensure generalizability and fairness, increase trust in detection models	Continuous improvement in data collection, annotation, and model optimization, balancing ethical considerations with accuracy and effectiveness

number. It accounts for true positives and negatives as well as false positives and negatives, providing a better picture of performance.

The AUC-ROC scores the ability of the model to tell apart each of the classes and thus gives us an understanding about what our model can do overall. The greater the AUC-ROC, the better will be the associated performance in discriminating hate speech from non-hate speech.

Inter-rater agreement – Kappa scores: These kappa scores are provided to assess the inter-rater agreement between two annotators, which is important for evaluating the consistency of annotations. This enforces alignment between human judgement and model predictions pinpointing areas of improvement for the model.

Along with these metrics, we also perform an error analysis looking at the linguistic level. Through this analysis, we explored misclassifications from dialectal differences, code-switching, and contextual nuances. This approach can enable targeted improvements by identifying the model weakness succinctly.

3.6 Hyperparameter-tuning, transfer learning and domain adaption

Hyperparameter tuning, which is one of the processes to improve machine learning models for hate speech detection. This is choosing the best configuration settings that control the learning process, such that predictive performance is maximized on a validation set (while still overfitting-free enough to ensure the model generalizes well to unseen data) (Bergstra and Bengio 2012).

Grid search, a very popular hyperparameter search and optimization method that defines a set of hyperparameters to find the best combination, and random search are commonly used methods for hyperparameter tuning. Bayesian optimization and other sophisticated methods can be used for efficient modelling of the performance function to give predictions on hyperparameters with strong potential.

This has made transfer learning techniques very effective in improving hate speech detection models. Transfer learning, allows us to use learned features from past experiences of training model(s) on very large datasets and applying on new tasks while retaining similarities. This is useful when labelled data is limited, expensive to obtain which is usually the case in hate speech detection. Fine-Tuning of ModelsThe existing data sets have quite the tendency to fine-tune large pre-trained language models (BERT or GPT-3) on specific hate speech datasets, which can improve the accuracy and robustness of these hate speech detection models [61].

Domain adaptation techniques are also important to guarantee model generalization across different contexts and datasets. This enables the model to generalise better by learning domain-invariant features and to adapt to specific

Table 14 Feature component comparison

Feature	2000s	2010s	2020s
Size and Type	Smaller, text-only (tweets, forum posts)	Larger, diverse (social media comments, news articles, blogs)	Massive, multimodal (text, images, videos)
Features	Simple (keywords, N-grams, sentiment)	Complex (linguistic features, user/network features, sarcasm detection)	Advanced (embeddings, emotional analysis, multimodal cues)
Strengths	Easy to collect, explicit hate speech focus	Context capture, support for advanced models, nuanced hate speech detection	Semantic understanding, real-world context, sarcasm/irony handling
Limitations	Limited variety, lack of context, poor irony/sarcasm handling	Data collection challenges, potential bias, labelling difficulty	High computational demands, multimodal annotation challenges, ethical concerns
Examples	HateBase, Sentiment140	Jigsaw Toxic Comment Challenge, Waseem and Abu-El-Haija (2018)	Multi-Domain Hate Speech Corpus, Hateful Memes Dataset
Challenges	Data scarcity, subjective labelling, limited context	Bias in user/network features, ensuring ground truth	Dynamic updates, cross-cultural diversity, explainable AI

linguistic and contextual characteristics of data across domains, respectively (Ruder 2019). Additionally, that creating synthetic examples so-called data augmentation can help improve the domain adaptation too. All of these strategies together help the hate speech detection model to be more generalized to different platforms and natural forms of online communication (Ganin et al. 2016).

3.7 Explainable AI and ethical considerations and bias Mitigation

Ethical Use of Data: The data collection and usage comply with ethical standards and privacy laws. Anonymize personal information to protect user privacy.

Bias Mitigation: It is ensured to work to identify and mitigate biases in the dataset and model. This includes biases related to gender, religion, caste, and regional dialects in the Hindi language.

Explainable AI (XAI) and Bias Mitigation in Hate Speech Detection are vital for ensuring transparency and fairness in AI systems.

Explainable AI (XAI): Explainable AI (XAI) techniques, such as LIME and SHAP, are essential for understanding hate speech detection behaviour of the AI model. LIME works by locally approximating the model (Ribeiro et al. 2016) to identify influential features of specific instances. SHAP (SHapley Additive exPlanations) values provide a single measure of feature importance derived from cooperative game theory (Lundberg and Lee 2017). These approaches promote transparency and ensure that decisions are based on legitimate reasons rather than discrimination or factors irrelevant to the decision.

Bias Detection and Mitigation: For instance, tackling gender-based bias, caste-related biases and regional biases are extremely important too. It includes following the principles of dataset bias that starts with balanced and representative datasets, use data de-biasing during pre-processing (Bellamy

et al. 2018) and fairness metrics such as demographic parity as well as equalized odds (Buolamwini and Gebru 2018). Subgroup analysis (identifies bias on specific group) To mitigate such biases, we can use algorithmic adjustments methods (e.g., re-weighting and adversarial debiasing or post-processing methods (e.g., equalized odds post-processing) (Hardt et al. 2016).

Continuous Monitoring and Feedback: These systems monitor ongoing model performance and flag biases as they arise. **Feedback Mechanisms:** This allows the users (and any affected communities) to report bias they have found, which helps with both model refinement and fairness improvement (Holstein et al. 2019). Incorporation of XAI principles in conjunction with the proposed strong bias detection and mitigation approaches can help in supporting a systemic approach to building transparent, fair and accountable hate speech detection systems that not only promote ethics but also social justice. Such steps will be an important part of AI systems earning societal trust for their value rather than risk protection and/or reinforcement of bias.

Ethical considerations play an important role in the design and implementation of hate speech detection systems. The potential for misuse, the privacy implications and the consequences of false positives and negatives highlight a powerful need for an ethics framework. To avoid abuse, there are explicit use policies, checks and balances in place to oversee these systems, and the ability for a user to challenge decisions. This keeps the system accountable and just for all. Anonymizing data, collecting only the required information, and securely storing and transmitting data all contribute to protecting user privacy. The adherence to data protection regulations like GDPR further ensures user privacy. Addressing false positives and negatives through a holistic approach. Like, Using Explainable AI (XAI) principles (state-of-the-art approaches such as LIME and SHAP), to provide insight on how the model is making decisions. It utilizes appropriately-balanced and diverse training data

to minimize bias and improve accuracy. This has allowed for continuous evaluation and refining of models, with a human-in-the-loop approach permitting review of final decisions. All of these measures, collectively, are intended to be implemented with a view to avoiding unfair, opaque and disrespectful hate speech detection systems that contravene ethical principles and undermine public trust. It is created with a view to encouraging constructive digital communication while also protecting individual rights and privacy.

3.8 Real-world challenges and advancements in research framework

However, implementing hate speech detection models in the real world involves several challenges. One major challenge is that it requires tremendous computation. However, implementing and maintaining tons of complex machine learning models require high processing power, memory, and storage; therefore expensive (Chollet 2017). Then there are complications with implementing such models into current systems. This is needed in order to operate a software that will be compatible with other environments, pipelines, data and user interfaces and performs according to the expectations without failure. Cross-lingual and multilingual models would be critical to making sure this is scalable across other Indic languages as well. Thus, these models generalize from language to language in a manner that the model trained on one native language can function adequately well on another foreign one after little retraining. This alleviates the requirement for massive annotated data for every single language and provides more stable performance (Ruder 2019). Transfer learning is one example where models trained on large corpora of a given language are applied and tuned to smaller datasets on another language. Additionally, some synthetic data generation strategies can be used to augment training datasets. As annotated data may not be available for many languages and in some specific hate speech context, these techniques deal with the lack of annotated data problem by generating synthetic instances similar to the real-world data. Creating a diverse set of synthetic examples allows models to become familiar with different variations and settings of hate speech making the overall performance better as well as more robust (Shorten and Khoshgoftaar 2019). Data augmentation, GANs (Generative-Adversarial-Network), synthetic oversampling is some generally used data enhancing methodologies. Mechanisms that allow users to provide feedback are crucial for improving model accuracy. When users flag something as hate speech when it isn't or vice versa, this feedback reveals weaknesses in the model. This creates a feedback loop that helps the model evolve with time, improving its accuracy. When users know how the flagging process works, it boosts their trust. It is therefore not enough to say that content has been flagged; users

must be able to read about the reasons for those flags in order to understand them. Explanations allow users to perceive the system as fair and trustworthy, which is essential for acceptance and effectiveness of an AI solution (Holstein et al. 2019).

Hybrid models of rule based and machine learning can generate better hate speech detection. Rule based systems learn what are patterns of hate speech while machine learning models recognise more subtle, context-dependent instances. Such combination exploits the advantages of both algorithms and produces a more robust detection system (Chowdhury et al. 2020). A rule-based component could catch explicit slurs, while a more context-dependent representation of hate speech (that would have an implicit usage) would be captured with the machine learning model. In addition, the anomaly detection techniques help in identifying Far right hate speech forms related to emergence. These are methods that identify unusual or anomalous behaviour within data, which differ from established forms of hate speech. The system will also be able to adapt as new patterns of hate speech language evolve, which will allow for it remain effective in the long-term, by the practice of continuous anomaly monitoring (Breunig et al. 2000). Clustering, density-based methods, and outlier detection algorithms are the techniques we can use for this. The combination of these developments improves both the strength and performance of hate speech detection systems, ensuring that they are flexible and robust in a variety of real-world situations. Overcoming computational costs, integration complexity, scalability, data scarcity, user feedback and emerging patterns can enable researchers and practitioners to design systems that are more robust and accurate allowing for safer online environments while also fostering a more diverse global community.

4 Research methodology and framework

4.1 Objective of bibliometric analysis and Prisma framework

Bibliographic analysis is a systematic method for examining and evaluating published research within a specific field or topic. It involves the collection, organization, and critical assessment of scholarly works, including journal articles, books, conference proceedings, and other relevant sources.

Mapping the Knowledge Landscape

Identify Key Concepts and Trends: Bibliographic analysis helps outline the major themes, theories, and debates within a field, revealing how knowledge has evolved over time.

Discover Research Gaps: Analyse what has already been thoroughly explored and pinpoint areas where further research is needed.

Tracing Intellectual Lineage: Understand the Flow of Ideas: Identify pivotal publications, influential scholars, and the connections between them. This helps to understand how scholarship has developed through citations and the evolution of research.

Recognize Seminal Works: Identify the publications that have significantly shaped a field, serving as foundational texts.

Appropriate Methodologies: Gain insights into the various research methods used in the field, their strengths, weaknesses, and suitability for different research questions.

Identify Suitable Datasets and Resources: Bibliographic analysis can point researchers to valuable datasets, repositories, and tools relevant to their work.

Position New Research Contributions: Understand how our own research fits within the broader landscape and its potential impact on the field.

Avoid Duplication of Effort: Ensure that our research question is original and hasn't already been sufficiently addressed by others.

Generate Review Papers: It is to produce a concrete overview of the current state of research, summarizing key findings and highlighting unresolved issues.

Support Meta-Analyses: Combine results from multiple studies to statistically analyse overall trends and identify robust findings.

PRISMA is a roadmap for conducting and reporting on high-quality research reviews. It's particularly focused on systematic reviews and meta-analyses, which are methods for combining and analysing results from multiple studies. Originally created for medical, PRISMA's principles can be helpful in many fields.

PRISMA Statement: This is the review's "report card." It includes:

- **Title:** Clearly states that it's a systematic review or meta-analysis.
- **Abstract:** A short summary of purpose, how we conducted the review, what is found, and what it means.
- **Introduction:** Why this review matters and the exact questions we aimed to answer.
- **Methods:** A detailed step-by-step explanation of the process, including how we searched for studies, decided which ones to include, analysed the data, and assessed for potential bias.
- **Results:** Presents what we found, including a breakdown of how many studies we included, their characteristics, and the combined results.

- **Discussion:** Explains the key findings, compares them to other research, highlights any limitations, and states overall conclusions.
- **Funding:** Acknowledges who funded the research.

PRISMA Advantages

- **Clear and Reproducible:** PRISMA makes sure that there is no miss in important details, so others could follow the same steps.
- **Trustworthy Research:** The structured approach helps minimize errors and biases, increasing the reliability of our conclusions.
- **Easy to Compare:** Using PRISMA makes it easier to compare different reviews on the same topic.
- **Planning Tool:** PRISMA can also guide the researchers to plan the research in right direction

4.2 Bibliometric analysis using R and VOS viewer

The initial phase of the research involved conducting a comprehensive bibliometric analysis to understand the current state of research in the field of hate speech detection, particularly focusing on the Hindi language. This involved:

- **Literature Review:** Systematic review of existing academic papers, journal articles, and conference proceedings from databases like IEEE Xplore, Google Scholar, and PubMed. The focus was on identifying prevailing methodologies, algorithms, and challenges specific to hate speech detection in Hindi.
- **Citation Analysis:** Examining the citation networks to identify influential studies, authors, and research trends in the domain of NLP and detection of hate speech.
- **Content Analysis:** Qualitative analysis of the literature to understand the scope, methodologies, and findings of previous research. This helped in identifying research gaps and emerging trends in hate speech detection.
- **The research begins with a bibliometric analysis, leveraging the capabilities of R, a powerful tool for data analysis and visualization. This phase involves:**
- **Data Retrieval:** Utilizing R packages like bibliometrix and scholar and VOS-viewer to systematically retrieve publications related to hate speech detection, particularly focusing on the Hindi language, from databases such as IEEE Xplore, Google Scholar, and PubMed.
- **Data Analysis:** Employing R and Vos-viewer's analytical capabilities to perform citation analysis, co-citation analysis, and content analysis. This includes identifying key authors, journals, and articles that have significantly contributed to the field, as well as emerging trends and research gaps.

- Visualization: Using R packages like ggplot2 and biblioshiny for visualizing data, which can include citation networks, trend analyses over time, and thematic mappings. These visualizations aid in better understanding the research landscape and the interconnections between different works (Fig. 4).

This study explores to classical ML/DL models along with the SOTA models, different hate speech datasets, and significant features of hate speech models, particularly focusing on the Hindi language. The methodology is inspired by the approach of Transfield et al. adapted to suit the specificities of hate speech detection in Hindi.

Thematic Map

See Table 15.

Parameter:

- field: "ID"—This likely indicates that the analysis or data processing is being done on some form of identifier field. It could be a unique identifier for each document or data entry.
- minfreq: 2—This suggests that the algorithm is set to consider only those terms that appear at least twice ('minimum frequency'). This is a common parameter in text analysis to filter out rare terms that may not contribute significantly to the pattern recognition.
- stemming: FALSE—Stemming is a process in NLP used to reduce words to their root form. Setting this to

'FALSE' indicates that stemming is not being applied, and the algorithm is considering the full form of words.

- n.labels: 3—This could refer to the number of labels or categories that the algorithm is supposed to identify or use in its classification. It might be related to the cluster assignments in the previous image.
- repel: FALSE—This parameter is less clear without context, but in some data visualization tools, 'repel' refers to whether or not labels in a visual representation are allowed to overlap. If 'repel' is set to 'FALSE', it may mean overlapping is allowed, which could make the visualization harder to read if there's a lot of data.

Analysis:

- Terms had to appear at least twice to be considered.
- Words were used in their full form, not stemmed to their roots.
- The analysis was set to categorize or label data into three distinct categories or clusters.
- Overlapping of labels in the visual representation of data (if any) was permitted.

Occurrences Map:

See Table 16.

Analysis Breakpoint

Here's a breakdown of what each column generally represents:

Fig. 4 Conducting bibliometric research using R

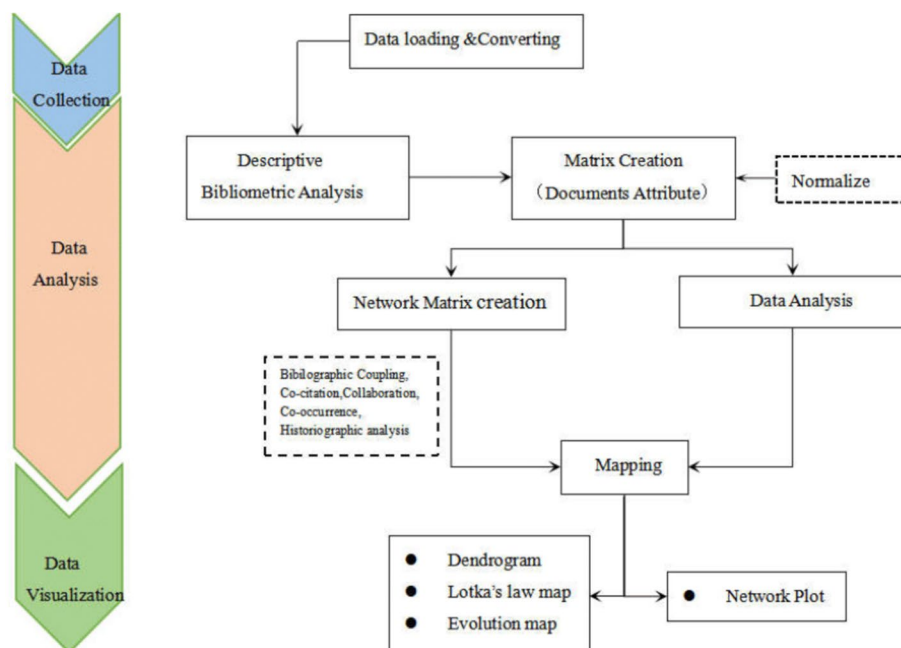


Table 15 Thematic map

Params	Values
Field	ID
Minfreq	2
ngrams	1
Stemming	FALSE
Size	0.3
n.labels	3
Community.repulsion	0
Repel	FALSE
cluster	walktrap

- Occurrences: How many times a term appears within the dataset or corpus.
- Words: Specific terms or keywords that were analyzed.
- Cluster: The grouping or category that the words are assigned to based on some criteria, likely through a clustering algorithm.
- Cluster_La: Possibly a measure specific to the clustering algorithm, which might indicate the significance of the term within its cluster.
- btw_centr: Betweenness centrality, a measure in graph theory that shows how often a node appears on the shortest paths between other nodes. In this context, it could indicate how central a term is within the network of terms.
- clos_centr: In the analysis of networks, closeness centrality serves as a valuable tool for determining a node's relative importance. Nodes with higher closeness centrality exhibit shorter average distances to all other nodes within the network. When applied to a textual dataset, this metric can reveal terms that hold central positions within the overall semantic structure of the text.
- pagerank: PageRank, a foundational algorithm employed by Google Search to assign ranking scores to web pages within their search results. In this context, 'importance' can denote a term's relative significance as determined by its

connections and relationships with other terms within the dataset

- Bibliometric Analytical observations:
- The terms are likely from a body of text related to AI research and have been processed to determine their relevance and relationships to each other.
- "Social network" (likely an abbreviation or typo of "social network") has the highest occurrence (5) in the 'speech recog' (likely "speech recognition") cluster and also the highest PageRank, which suggests it's a very relevant term in this context.
- "Classification" appears in two different clusters with varying centrality measures, suggesting it might be a bridging term connecting different areas of AI.
- The "deep learning" cluster has fewer occurrences of terms but each with a reasonably high PageRank, indicating that while the terms may not be as frequently mentioned, they're still considered quite significant.
- The btw_centr and clos_centr measures seem to correlate with the number of occurrences, but not perfectly. This could imply that some less frequently occurring terms are still central within the network of terms.
- The Cluster_Lambda (Cluster_La) column's meaning is not immediately clear without additional context, but it might indicate the strength or coherence of the cluster (Table 17).

Following breakdown of the metrics provided:

CallonCent: Likely a centrality measure based on the methodology of Michel Callon; a sociologist known for his work on scientometrics. The numbers (1.27777778 for classification and 2.5 for deep learning) suggest that 'deep learning' has a higher centrality within its network compared to 'classification (of information)'.

CallonDen: This could stand for Callon's density, another scientometric measure that indicates the strength of the ties within a cluster. A higher number (63.8888889 for

Table 16 Occurrences map

Occurrences	Words	Cluster	Cluster_Label	btw_centr	clos_centr	pagerank
3	Classification	1	classification	686.452535	0.00552486	0.01695958
2	Benchmarking	1	classification	418.607929	0.00510204	0.0131011
5	Social network	2	speech recog	1070.0103	0.00636943	0.0305207
4	Learning syste	2	speech recog	326.776287	0.00546448	0.02346094
3	Computatione	2	speech recog	181.123343	0.00549451	0.01839819
3	Hate speech d	2	speech recog	234.892445	0.00515464	0.01721293
2	Convolutional	2	speech recog	140.805237	0.00546448	0.01422678
2	Modeling lang	2	speech recog	51.2378156	0.0049505	0.01288622
2	Speech comm	2	speech recog	61.263469	0.00505051	0.01448699
2	Language proc	3	deep learning	36.0435239	0.00507614	0.01180792
2	Natural languag	3	deep learning	36.0435239	0.00507614	0.01180792

classification and 100 for deep learning) implies that the 'deep learning' cluster is more tightly connected.

RankCentr: This could be a ranking based on centrality, where 1 is the highest rank. Both clusters have a value of 1, which might indicate they are equally ranked at the top in their respective contexts, or that they are the only clusters being considered (hence both ranked as 1).

RankDensi: Similar to RankCentr, this is likely a rank based on density. Both clusters are also ranked as 1, suggesting that they might be the most densely connected clusters within their respective networks or that they are the sole focus of the analysis.

ClusterFreq: This indicates the frequency of the clusters within the dataset or corpus. 'Classification (of information)' appears 7 times, while 'deep learning' appears 10 times. This suggests that 'deep learning' is discussed more frequently.

Analysis:

- **Deep Learning Cluster:** This cluster appears to be more central (higher CallonCent) and more densely connected (higher CallonDen) within its network. It also has a higher occurrence in the dataset (ClusterFreq of 10), indicating that it may be a more dominant or focused-upon topic within the body of text or research analyzed.
- **Classification Cluster:** Despite a lower centrality and density score, it still ranks equally with the deep learning cluster in terms of RankCentr and RankDensi, which might imply that both clusters are of significant interest in the analyzed data.

The overall analysis suggests that the 'deep learning' cluster is perhaps a more cohesive and frequently occurring topic than 'classification (of information)', but both are considered primary clusters based on their rank scores (Fig. 5).

Following are some observations based on the network graph:

- **Clusters:** The graph has several clusters, each likely representing a group of keywords that are frequently mentioned together. Each cluster seems to be color-coded to differentiate between them.
- **Central Keywords:** Some keywords, like "social networking (online)", "speech recognition/speech detection", and "deep learning", are more central and have many connections, suggesting they are common co-occurring terms

within the dataset. They likely serve as key topics or focal points in the dataset.

- **Connections Between Clusters:** There are lines connecting different clusters, which indicate that there are some co-occurrences between keywords in different categories. For instance, "classification (of information)" has connections to both "social networking (online)" and "benchmarking", indicating a relationship or shared context in the dataset.
- **Isolated Subgraphs:** Some smaller subgraphs are less connected to the main clusters, like "behavioural research" and what appears to be "decision matrix" and "decision support". Their isolation from the main clusters could mean they are niche topics within the dataset or discussed in a more specific context.
- **Density and Centrality:** The graph shows varying densities within clusters, with "social networking (online)" having a particularly dense network, which correlates with the CallonDensity value seen in the previous table. This supports the idea that "social networking (online)" is a key topic with many interrelated terms. The centrality of nodes like "deep learning" in the graph also aligns with the high CallonCentrality score provided earlier, confirming its importance in the dataset.
- **Edge Thickness and Node Size:** Thicker edges might represent more frequent co-occurrences, and larger node sizes could indicate higher occurrences or centrality measures. However, without specific legends or scales, this is speculative.

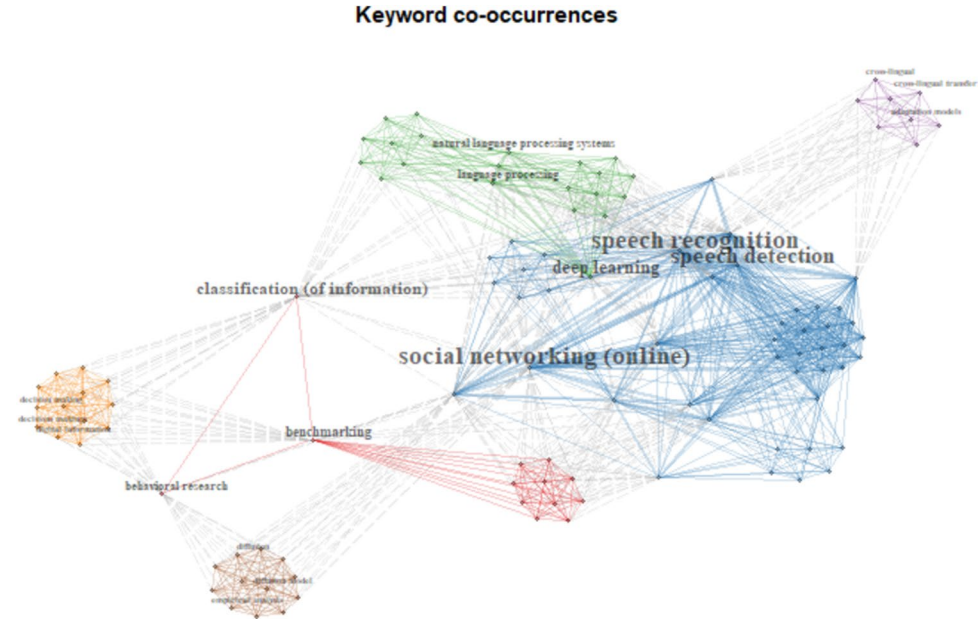
See Table 18.

- **field: "ID"**—This suggests that the analysis is being conducted on some form of data identified by an "ID" field, which is common in databases or datasets where each entry needs to be unique.
- **method: "MCA"**—This refers to Multiple Correspondence Analysis, a statistical technique used to analyze and visualize the patterns of categorical data. It can help in detecting and interpreting underlying relationships between different categories of data.
- **clust: 1**—This parameter could be specifying that only one cluster should be identified in the analysis. This might be in contrast to other settings or analyses where multiple clusters are defined.

Table 17 Cluster observation

Cluster	CallonCent	CallonDen	RankCentr	RankDensi	ClusterFreq
Classification (of information)	1.27777778	63.8888889	1	1	7
Deep learning	2.5	100	2	2	10

Fig. 5 Keyword co-occurrence network



- stemming: FALSE—This indicates that stemming, a NLP in method that reduces words to their base or root form, is not applied in the analysis.
- documents: 5—This parameter indicates the number of documents included in the analysis. It could be a filter to consider only a subset of the data, or it could indicate that the dataset is composed of 5 documents.

Interpreting these parameters in context.

- The MCA method suggests that the analysis is focused on exploring the relationships between categories within the dataset, possibly to understand how various terms or items are associated with each other.
- The 'clust' value of 1 could imply that the analysis is not meant to segment the data into multiple groups but rather to treat it as a single entity, which might be unusual for clustering but could be relevant for certain types of MCA.
- The 'documents' value shows that only a small set of documents is being analyzed, which might mean that this is either a pilot study, a small-scale exploratory analysis, or a demonstration of the method.
- The decision not to apply stemming means the analysis is considering the full lexical form of words, which could be important for understanding the full context or for maintaining the specificity of technical terms.

See Table 19.
Following is an analysis breakdown:

Table 18 Bibliometric parameters

Params	Values
Field	ID
rgrams	1
Method	MCA
Clust	1
k.max	8
Stemming	FALSE
labelsiz	5
Documents	5
graph	FALSE

- word: This column lists different terms that are likely extracted from a dataset, possibly related to topics in natural language processing, machine learning, and data science.
- Dim_1 and Dim_2: These columns show the coordinates of each word on two principal dimensions that result from the MCA. The values represent how each term is positioned in the reduced dimensional space, which helps to visualize and interpret the relationships between terms.
- cluster: All words are assigned to cluster '1', which indicates that in this particular analysis, all terms are considered part of a single group or cluster. This aligns with the parameter 'clust: 1' we provided earlier.

Analysing the dimensions:

- Words with similar coordinates on both dimensions are likely to be related or used in similar contexts within the dataset. For instance, words with high values on both dimensions, like "ensemble.deep.learning" and "cyberbully.detection", may share a thematic or contextual connection.
- Conversely, words with opposing values on a dimension might represent different or opposing concepts or themes. For instance, "deep.learning" has a negative value on Dim_1, while "modelling.languages" has a positive value, suggesting a potential divergence in context or usage.
- The positioning on Dim_1 and Dim_2 helps to understand the latent structures in the data, revealing how terms are associated across the dataset. For example, terms like "bi.directional.gated.recurrent.unit" have high positive values on both dimensions, indicating a strong presence or significance in the analysed context.
- It's worth noting that the meaning and interpretation of these dimensions depend on the dataset and the exact method used for analysis. Without additional context, we can only speculate on the thematic relationships based on the positioning.

The clustering assignment to '1' for all terms suggests a single category or topic is being analysed, or it could be a simplification for illustrative purposes (Fig. 6).

Following we analyse on the visual information.

- **Axes:** The horizontal and vertical axis signify the two dimensions of the reduced space onto which the data points (words or phrases) are plotted. The origin (where Dim_1 and Dim_2 intersect) is the point of average values for the dimensions.
- **Data Points:** Each data point represents a word or phrase from the dataset. The proximity of points to each other indicates how closely related they are in the dataset's multidimensional space.
- **Quadrants:** The plot is divided into four quadrants by the axes. The positioning of the points within these quadrants can sometimes suggest different thematic groupings or relationships.
- **Thematic Groupings:** Words that are plotted close to each other might be used together more frequently or share similar contexts within the dataset. For example, "deep learning" is close to "natural language processing systems" and "language processing", suggesting a relationship between these topics.
- **Outliers:** Points that are farther away from the origin and other data points could be considered outliers. They represent words or phrases that have a unique place in the dataset relative to others. For example, "benchmarking" and "behavioral.research" are relatively far from the clus-

Table 19 Keyword cluster association

Word	Dim.1	Dim.2	Cluster
Deep learning	-0.16	-0.82	1
Speech recognition	0.33	-0.49	1
Social networking online	0.27	-0.18	1
Speech detection	0.47	-0.4	1
Learning systems	0.58	-0.44	1
Social media	-0.29	-0.32	1
Classification of information	-0.59	0.18	1
Computational linguistics	0.85	0.33	1
f1scores	-0.14	-0.11	1
Hate speech detection	0.88	0.05	1
Long short-term memory	0.83	-0.13	1
Behavioural research	-0.89	-1.24	1
benchmarking	-0.62	1.24	1
Convolutional neural networks	1.24	0.16	1
Hate speech	-0.17	-0.47	1
Language processing	-0.2	-1.26	1
Modelling languages	1.36	-0.27	1
Natural language processing systems	-0.2	-1.26	1
Natural languages	-0.2	-1.26	1
semantics	-0.06	-0.12	1
Speech communication	1.33	-0.42	1
Adaptation models	-0.06	-0.38	1
Aggregation attention	0.05	-0.22	1
Attention head	0.05	-0.22	1
Automatic detection	-0.18	0.45	1
Bidirectional	2.66	0.75	1
Bi directional gated recurrent unit	2.66	0.75	1
Bi directional long short term memory	2.66	0.75	1
Bismuth compounds	-0.18	0.45	1
brain	2.66	0.75	1
Computer simulation languages	2.66	0.75	1
convolution	-0.18	0.45	1
Convolutional neural network	-0.18	0.45	1
Cross lingual	-0.06	-0.38	1
Cross lingual transfer	-0.06	-0.38	1
cyberbully	2.66	0.75	1
Cyberbully detection	2.66	0.75	1
Data labelling	-0.41	-0.93	1
Data stream	2.66	0.75	1
Decision making	-1.06	2.04	1
Decision trees	0	-1.59	1

ter of other terms, which could indicate a distinct context or less frequent co-occurrence.

- **Cluster Shape:** The overall shape of the cluster of points can provide insights as well. In this plot, there is a wide dispersion along the horizontal axis compared to the ver-

tical axis, which could suggest that Dim_1 explains a greater variance in the dataset than Dim_2.

- **Interpretation of Dimensions:** The exact meaning of Dim_1 and Dim_2 is not inherently clear without further context. They are derived from the data through the MCA and represent the most significant variations in how words or phrases co-occur within the dataset.

See Table 20.

- **Occurrences:** Indicates the number of times each term appears in the dataset. For example, "classification" appears three times, while "benchmarking" appears twice.
- **Words:** This column lists specific terms that were analysed. They seem to be truncated, likely due to space constraints in the table (e.g., "classification", "benchmarking", "social network").
- **Cluster:** The grouping or category that the words are assigned to, which in this case, there are three distinct clusters.

- **Cluster Label:** Provides a label for each cluster, which gives context to the type of terms grouped within it. Cluster 1 is labelled "classification (of information)", Cluster 2 is "speech recognition", and Cluster 3 is "deep learning".
- **btw_centrality (Betweenness Centrality):** This signifies the amount of centrality in a graph based on shortest paths. For any given term, it represents the number of times it acts as a bridge along the shortest path between two other terms. For instance, "classification" has the highest betweenness centrality, indicating it plays a significant role in connecting different terms or concepts within the network.
- **clos_centrality (Closeness Centrality):** Reflects how close a term is to all other terms in the network. A higher closeness centrality means that the term is, on average, shorter distances to all other terms. For example, "classification" again has a high closeness centrality, reinforcing its importance in the dataset.
- **pagerank_centrality (PageRank Centrality):** Derived from the PageRank algorithm, this metric indicates the relative significance of each term within the network.

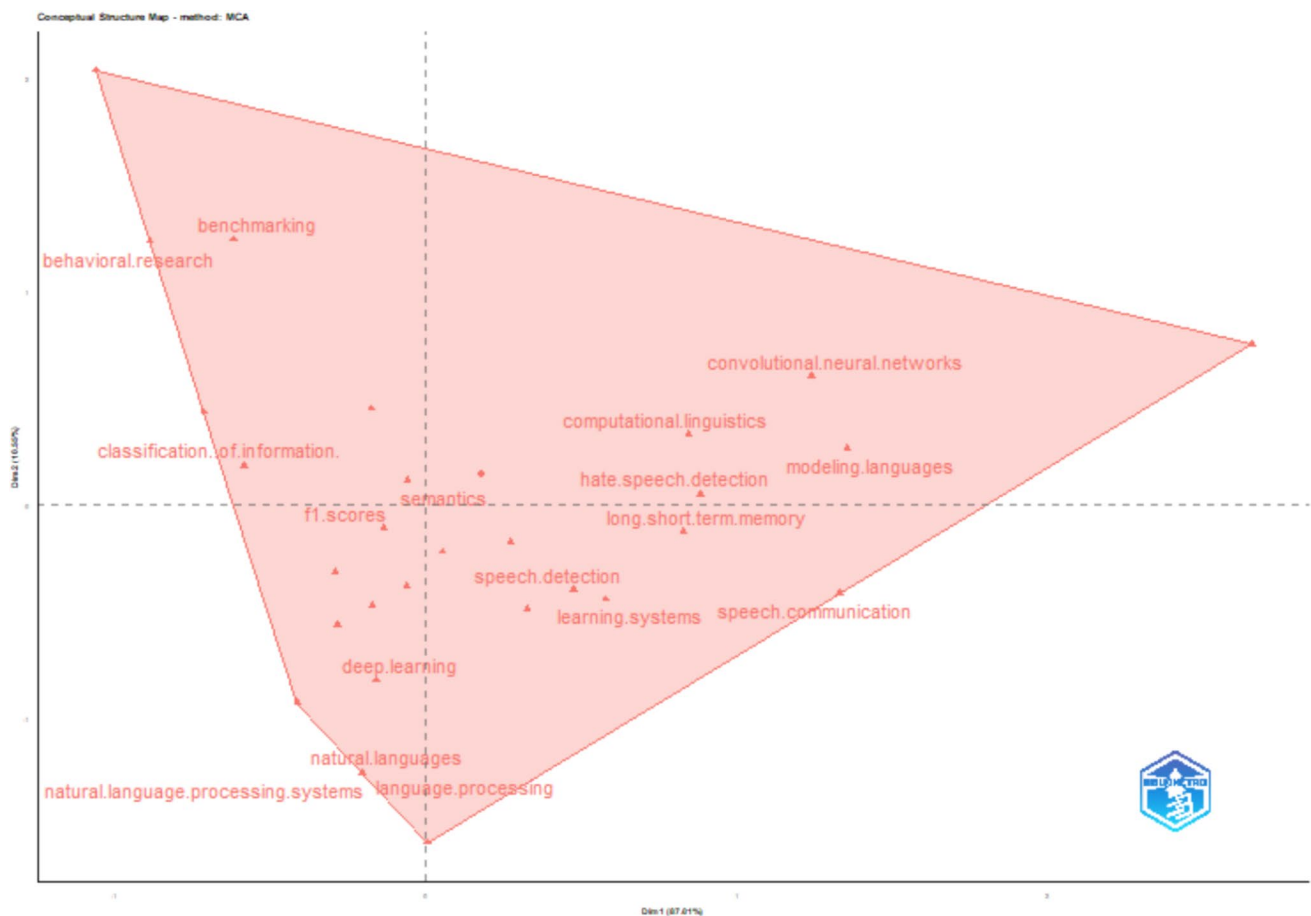


Fig. 6 Cluster mapping of keywords

A higher PageRank suggests that a term is more important or better connected. "Classification" has the highest PageRank, aligning with its central role suggested by the other centrality measures.

Analysis of the clusters

- Cluster 1: Contains terms related to the classification of information, which seems to be a central theme in the dataset, as indicated by the high centrality measures.
- Cluster 2: Focused on speech recognition, it has the most terms listed and includes a variety of concepts related to speech and language processing.
- Cluster 3: Contains terms related to deep learning. It's not as densely populated as Cluster 2, but the terms have a reasonable closeness and PageRank centrality, indicating their significance (Fig. 7).

This type of diagram maps the research themes or topics according to two dimensions:

Horizontal Axis (Relevance degree/Centrality): Measures the degree of interaction of a theme with other themes within the network. Themes with higher centrality are considered 'basic' and are fundamental to the structure of the research field, while themes with lower centrality are more peripheral.

Vertical Axis (Development degree/Density): Represents the strength of the internal links between the keywords within a theme. Themes with higher density are well developed and have strong internal ties, whereas themes with lower density are less developed.

Based on these axes, the diagram is divided into four quadrants:

- Upper Right (Motor-Themes): Topics in this quadrant have both high centrality and high density. They are well-developed and important within the field. In this image, "speech recognition", "social networking (online)", and "speech detection" are considered motor-themes, indicating they are currently major areas of focus and development in the field.
- Upper Left (Niche Themes): The depiction of these segments with high density but low centrality. They are well structured in their own right but are not central to the research field. The absence of terms in this quadrant in the provided image suggests there are no niche themes, or they were not included in this visualization.
- Lower Right (Basic Themes): Themes here have high centrality but low density, indicating they are important to the field but not yet fully developed. "Deep learning", "language processing", and "natural language processing systems" fall into this category, suggesting they are emerging or basic themes that are central but less internally cohesive.
- Lower Left (Declining or Emerging Themes): These are segments with both low centrality and low density. They could be either declining in importance or just emerging. "Classification (of information)", "behavioural research", and "benchmarking" are placed here, indicating they might be less focused areas or possibly declining in relevance (Fig. 8).

Observations

- Each bar represents an individual author, with the authors' last names and initial(s) listed along the vertical axis.
- The horizontal axis represents the H-index value, ranging from 0.00 to 1.00. This is unusual because H-index values are typically whole numbers. The scale here suggests

Table 20 Words clustering score

Occurrences	Words	Cluster	Cluster_Label	btw_centrality	clos_centrality	pagerank_centrality
3	classification	1	classification (of information)	686.452535	0.00552486	0.016959582
2	benchmarking	1	classification (of information)	418.607929	0.005102041	0.013101104
5	social network	2	speech recognition	1070.010301	0.006369427	0.030520702
4	learning system	2	speech recognition	326.776287	0.005464481	0.023460938
3	computation	2	speech recognition	181.123429	0.005494505	0.018398193
3	hate speech d	2	speech recognition	234.892445	0.005154639	0.017212933
2	convolutional	2	speech recognition	140.805237	0.005464481	0.014226781
2	modeling lang	2	speech recognition	51.23781556	0.004950495	0.012886222
2	speech comm	2	speech recognition	61.26346901	0.005050505	0.014486989
2	language proc	3	deep learning	36.04352392	0.005076142	0.011807916
2	natural language	3	deep learning	36.04352392	0.005076142	0.011807916

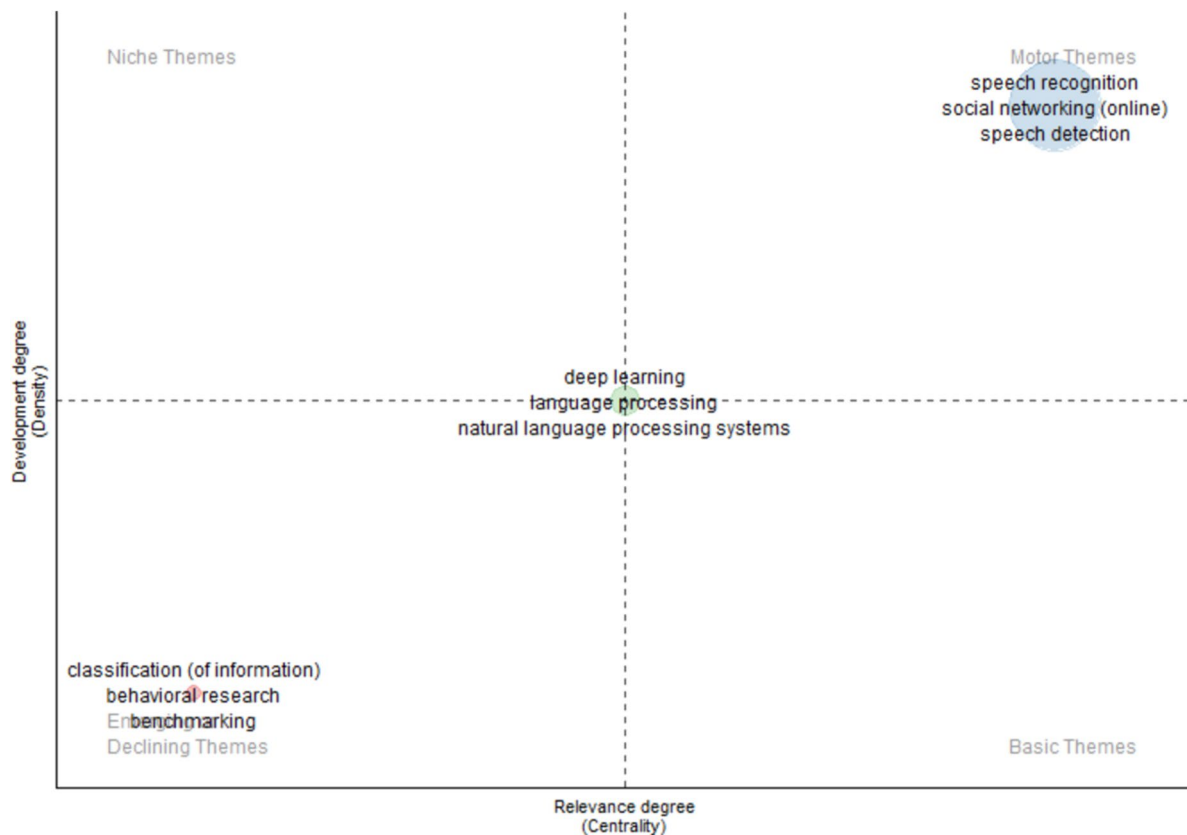


Fig. 7 Thematic placement of keywords

either a normalized or relative measure of the H-index rather than the absolute H-index value, or possibly an H-index adjusted to a 0–1 scale for comparison purposes.

- All authors shown have the same length of the bar, reaching 1.00 on the horizontal axis. This uniformity suggests that within the local context or the specific dataset being analysed, each author has a maximum impact value or they all share the same H-index value.
- The uniformity could also indicate that the dataset or community is very homogenous in terms of citation impact or that a threshold was applied where only those with the highest H-index were included.
- There is no differentiation between the authors in terms of their impact, so based on this graph alone, we cannot rank the authors or determine any differences in their scholarly impact (Fig. 9).

From this network visualization, we can infer the following:

- There is a significant amount of research activity or discussion related to "deep learning" and "speech recognition", as indicated by the central positioning and size of these nodes.

- "Computational linguistics" and "modelling languages" form another distinct cluster, indicating a strong thematic link between these topics.
- "Social networking (online)" and "social media" are closely related, which is expected given the overlap in these fields.
- The connection patterns suggest interdisciplinary research, with terms bridging different clusters.

4.3 Distribution of hate speech research paper through years

The distribution of hate speech research papers has seen a significant increase over the years, particularly in the wake of the emerge of social media and the growing awareness of online hate speech. Following a breakdown of the trend (Fig. 10).

Early years (pre-2000s)

Hate speech research was relatively limited, with a focus on legal and philosophical frameworks.

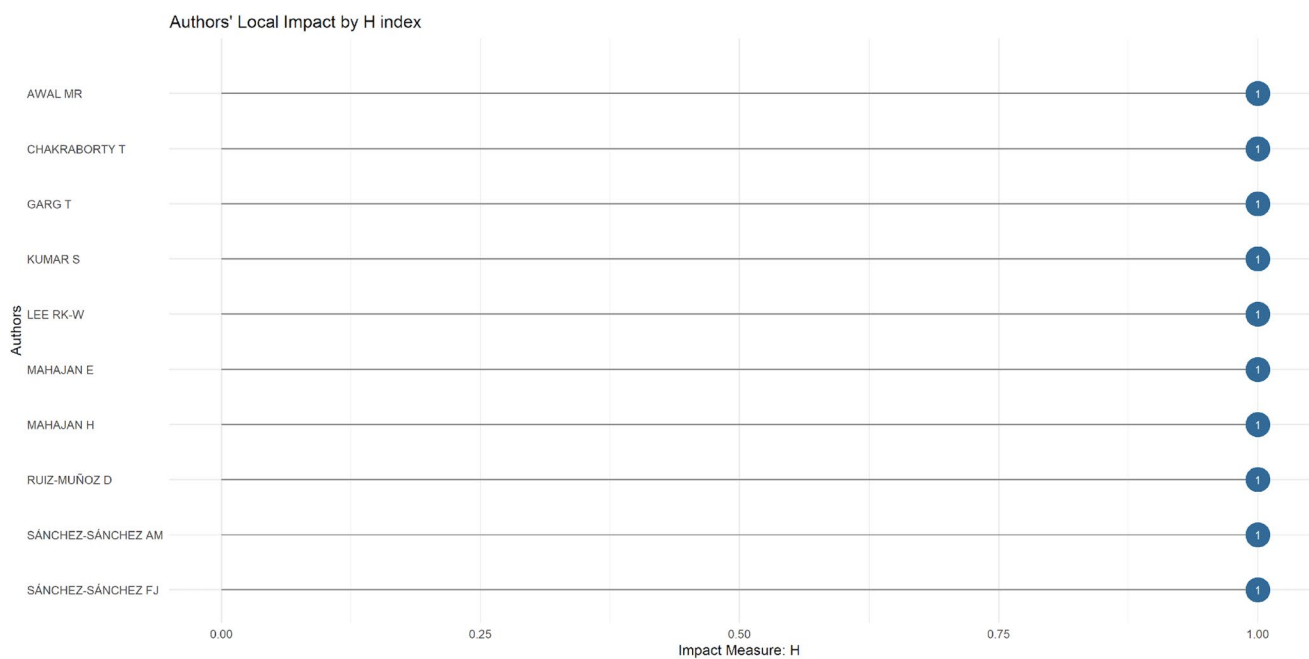


Fig. 8 H indexing impact

The number of publications was low, with studies scattered across various disciplines like law, linguistics, and sociology.

2000s

The rise of the internet and online hate speech led to a surge in research interest.

Studies began to focus on analysing online hate speech content, developing detection methods, and understanding its psychological and social impacts.

2010s

The penetration of social networking sites like Twitter and Facebook further amplified the issue of online hate speech.

Research efforts intensified, with a focus on:

Automatic detection and classification of hate speech: This involved developing machine learning algorithms to identify hateful content based on textual and linguistic features.

Impact of hate speech: Studies explored the psychological harms of hate speech on both victims and perpetrators, as well as its implications for social cohesion and democracy.

Countermeasures and interventions: Researchers investigated strategies to combat hate speech, including platform regulation, content moderation, and educational campaigns.

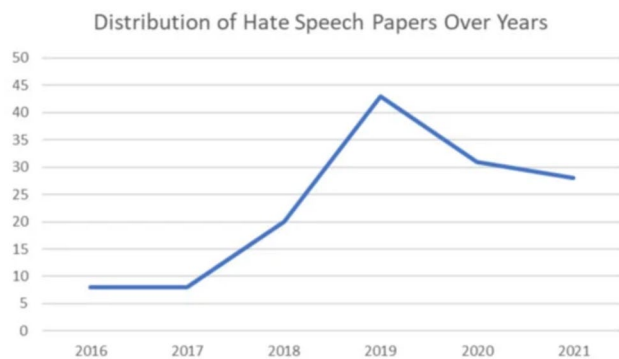
2020s

The trend of increasing research continues, with even more sophisticated and nuanced approaches being adopted. Some key areas of focus include:

Contextual understanding of hate speech: Moving beyond just textual analysis, researchers are now considering the broader context in which hate speech occurs, including factors like speaker intent, audience reaction, and platform norms.

Multilingual and cross-cultural perspectives: With hate speech prevalent across languages and cultures, research is expanding to understand its diverse manifestations and develop effective countermeasures in different contexts.

Ethical considerations: The use of automated technologies for hate speech detection raises concerns about bias, freedom of expression, and privacy. Research is addressing these ethical challenges to ensure responsible development and deployment of such technologies.

Fig. 9 Connection network**Fig. 10** Distribution of hate speech. Source: <https://www.mdpi.com/2078-2489/13/6/273>

4.4 Absence of fine-grained classification methods

Current hate speech detection research primarily uses binary classification, categorizing content as either "hate speech" or "non-hate speech." However, this binary approach fails to capture the complexity of hate speech, which ranges from explicit threats to subtle biases and micro-aggressions. There is a need for more nuanced models that can classify different

levels and types of hate speech, providing deeper insight and enabling more effective moderation (Davidson et al. 2017).

4.4.1 Focus on Hindi Language

Despite the global prevalence of Hindi and its linguistic complexities such as dialectal variation, code-mixing, and script diversity—research on hate speech detection in Hindi remains limited. Given the widespread use of Hindi on social media, there is a critical need for focused research to bridge this gap and develop effective hate speech detection models tailored to the language (Bohra et al. 2018).

4.4.2 Bias mitigation

Machine learning models trained on biased data can produce unfair or discriminatory results. There is a significant lack of research dedicated to systematically mitigating bias in hate speech detection models. Developing strategies to address this issue is essential to ensure models are both accurate and equitable, preventing the reinforcement of social biases (Sap et al. 2019).

4.4.3 High-quality multilingual datasets

The scarcity of high-quality, labelled datasets in languages other than English is a major obstacle to the development of effective hate speech detection models. In multilingual contexts, particularly for languages like Hindi, this lack limits the generalizability and robustness of these models. The creation of extensive multilingual datasets is urgently needed to support the development of more sophisticated and universally applicable hate speech detection models (Waseem et al. 2016).

5 Conclusion

5.1 Conclusion

Hate speech detection in the Hindi language setting highlights challenges and required solutions for building effective, aware-of-culture, and understanding-in-depth AI systems. We have covered the wide-ranging complications involved in this task, from linguistic variety and script differences to sociolinguistic contexts that separate hate speech from other types of communication.

The interplay of those Hindi colloquialisms, regional slangs, and code-switching is highlighted as a significant challenge to automated systems throughout the literature. Nevertheless, the field of NLP and ML has evolved further towards complex models that can identify trends and contexts more effectively. Contextual embeddings, transfer learning, and sentiment analysis are some of the techniques that have improved hate speech detection algorithms and provided a better understanding of the intent and potential harm behind online content.

The review also emphasizes the importance of datasets being comprehensive and balanced. However, a major roadblock came in the form of a lack of availability of labelled data for hate speech detection in Hindi. We are already loose on tackling this challenge due to creative algorithms like synthetic data and semi-supervised learning but opened up a fairly new space via data augmentation of which the users could eventually partake. In addition, native speakers and subject matter experts also take part in data annotation process to ensure that the content has cultural relevance and therefore enhances models' reliability.

In addition to this, finally the ethical aspect and considerations of hate speech detection cannot be ignored. It is a fine line between refraining or prohibiting hate speech from our society while at the same time ensuring that free speech is not being compromised. Future efforts should be directed towards developing AI systems that are transparent, accountable, and equitable in a progressive manner. We must also continuously monitor them for bias and second-order effects,

keeping them safe from hate speech but still upholding the principles of equity and fairness.

The field of Hate Speech detection in Hindi, not only embodies the potentials of AI creating a safer internet for all, but also shows us the heavy price that comes with it. This will finally be driven by the collective action of researchers, practitioners and policymakers. Using cutting-edge technologies, promoting inter-disciplinary collaboration and dictating high ethical standards makes one have a hope that the entire damaging effect of spreading hate speech can somehow be diminished and more integrated into the trend of dignifying and embracing digital communication.

5.2 Future scope and potential challenges

In recent years, a considerable amount of work has been focused on the detection of hate speech across multiple languages. This research will specifically build machine learning and deep learning models for the Hindi language, while ensuring that all the unique problems including code-switching, geographical dialects and script variations are solved for their respective datasets. However, even though Hindi is among the most widely used languages in the world, research work on hate speech detection has not received as much attention in English as it has. The purpose of this project is to develop the best possible models for detecting and categorizing hate speech in Hindi (Solorio et al. 2014).

Hate speech detection algorithms find code-switching and multilingual contexts a major bottleneck. Since mixed-language content does not follow a standardized spelling and grammar due to word borrowing (Barman et al. 2018), existing models trained on monolingual datasets cannot be applied (Agarwal and Kumar 2021). To address the diversity of languages and cultural differences, rich multilingual corpora and NLP models are needed (Chakravarthi et al. 2020).

There are several aspects of interest here Hindi NLP models, datasets, code-mixing, cultural/contextual understanding and so on. The current work would not only help advance hate speech detection for Hindi, but would also aid in cross-lingual efforts towards hate speech detection (Bohra et al. 2018; Davidson et al. 2017).

Future Scope: Future datasets may also be cross-linguistic and multi-modal – image, videos, audio and so on to better track hate speech dissemination through different forms of media. Such improvements may result in stronger detection systems (Matamoros-Fernández 2017). Also, future work should focus on the socio-cultural consequences of hate speech and the need to build real-time detection systems (Waseem and Hovy 2016).

Challenges Faced by Hindi Hate Speech Detector: The dialect diversity, code-switching phenomenon, and lack of data availability for multiple stages among Hindi linguistic circles

make it difficult to provide adequate work from NLP models. The perspective of the detected entity become bias in machine learning model (Zou and Schiebinger, 2018; Lipton, 2020) and cultural context (Giu et al. 2019), sarcasm and homographs (Gladkova et al. 2017), Slang and political sensitivities introduce additional layers of complexity to the challenge of detecting hate speech detection (Sap et al. 2019; Joshi et al. Making sure our algorithms are fair and lawful can be complicated, depending on the legal definitions of hate speech vary from state to state (Zarsky 2016).

Author contribution Each of the author have equally contributed to the research.

Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

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